CHAMBER FERMION LOCALITY AND LOCAL INTEGRATION

Abstract. We prove uniform locality estimates for penalized chamber Dirac operators on three-dimensional periodic lattices. A gap on the mean-zero chamber components and a free Schur-complement estimate give local inverse bounds independent of the mesh and reference mass. Local real gauges and convergent walk expansions extend these bounds to complex connections. We obtain analytic coarse covariances, an exact weighted Grassmann block identity, and a local expansion of the normalized determinant. Under additional charge and volume conditions, the local determinant difference gains a factor $e_k^{1/4}$.

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1. Introduction

Fix $0 < \rho < 1$, an odd integer $L \geq 5$, $0 < \alpha < 1$, and $\varepsilon > 0$. At level $k \geq 0$, put $\eta = \eta_k = L^{-k}$. A photon site is $c \in \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$; a chamber site is $x = (c, g)$, where $g \in O_{\mathrm{cub}}$ is a signed permutation matrix. There are 48 labels g and four spin components at each photon site. Connections are periodic complex oriented bond functions, with $A(c', c) = -A(c, c')$; spinors are antiperiodic in each coordinate. The chamber measure is $\eta^3/48$, while the photon-site spinor measure is η^3. Operator kernels use the chamber measure in their column variable; the identity kernel is $48\eta^{-3}\mathbf{1}[x = x']1_4$. The cluster injection and mean are

$$(J\xi)(c, g) = \xi(c), \qquad J^* f(c) = 48^{-1} \sum_g f(c, g).$$

Adjoints use these Hermitian pairings. Transposes use the corresponding bilinear pairings, without complex conjugation.

Use the Pauli and Dirac matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \gamma_j = \sigma_3 \otimes \sigma_j, \quad \gamma_5 = \sigma_1 \otimes 1_2.$$

Let s_0 reverse the third coordinate, s_1 interchange coordinates $2, 3$, and s_2 interchange $1, 2$; let r_{e_1} reverse the first coordinate. The four neighbours, inward normals, and weights are

$$x^{(i)} = (c, gs_i) \quad (0 \leq i \leq 2), \qquad x^{(3)} = (c + \eta g e_1, g r_{e_1}),$$
$$(n_0, n_1, n_2, n_3) = (e_3, (e_2 - e_3)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}, -e_1),$$
$$(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6).$$

For a charge $e_k > 0$, the chamber operator is

$$(D_A f)(x) = -\eta^{-1} \sum_{i=0}^{3} \frac{w_i}{2} (\rho + \gamma \cdot g n_i)(U_A(x, x^{(i)}) f(x^{(i)}) - f(x)), \tag{1}$$

where internal transports are one and the external transport is $\exp(ie_k \eta A(c, c + \eta g e_1))$.

The unit blocks are $\Delta_z = \{c : |c - z|_\infty < 1/2\}$, $z \in \mathbb{Z}^3/\ell\mathbb{Z}^3$. For $c \in \Delta_z$, let $c = c_0, c_1, \dots, c_k = z$ be the centres of its nested blocks of sides $L^j\eta$. Write $\tau A(u, v)$ for the arithmetic mean of the oriented fine-bond sums along the six coordinate-order paths from u to v. Set

$$\varphi_A(z, c) = \exp\left(ie_k \eta \sum_{j<k} \tau A(c_{j+1}, c_j) \right), \quad (Q_k(A)\xi)(z) = \eta^3 \sum_{c \in \Delta_z} \varphi_A(z, c)\xi(c),$$
$$(Q_k^T(-A)v)(c) = \varphi_A(z, c)^{-1} v(z).$$

For $z' \in L\mathbb{Z}^3/\ell\mathbb{Z}^3$, put $B(z') = \{z : |z - z'|_\infty < L/2\}$. The next average and its algebraic reverse are

$$U_A(z', z) = \exp(ie_k \eta \tau A(z', z)),$$
$$(Q(A)v)(z') = L^{-3} \sum_{z \in B(z')} U_A(z', z) v(z), \quad (Q^T(-A)v')(z) = U_A(z', z)^{-1} v'(z').$$

The target pairing has measure L^3. These maps act componentwise in spin. For $k \geq 1$, define

$$\Pi_1(A) = JQ_k^T(-A)Q_k(A)J^*, \qquad \Pi_L(A) = JQ_k^T(-A)Q^T(-A)Q(A)Q_k(A)J^*.$$

At $k = 0$ use only $\Pi_L(A) = JQ^T(-A)Q(A)J^*$. Let $b = 1$, $b_L = L^{-1}$, $b_- = 1 - L^{-1}$, and

$$b_k = \frac{1 - L^{-1}}{1 - L^{-k}} \quad (k \geq 1), \qquad \alpha_y = \frac{b_k i\gamma_5 y}{b_k + i\gamma_5 y}, \qquad \beta_y = \frac{b_k^2 b_L}{(b_k + b_L + i\gamma_5 y)(b_k + i\gamma_5 y)}.$$

At $y = \infty$, put $\alpha_\infty = b_k$, $\beta_\infty = 0$. For $0 < \mu \leq 1$, the penalized operators are

$$F_y(A) = \begin{cases} D_A + \mu + \alpha_y \Pi_1(A) + \beta_y \Pi_L(A), & k \geq 1,\ y \in \mathbb{R} \cup \{\infty\}, \\ D_A + \mu + b_L \Pi_L(A) + i\gamma_5 y, & k = 0,\ y \in \mathbb{R}. \end{cases} \tag{2}$$

The data k, μ, ℓ, e_k are independent.

Let $M \geq L^2$ be a power of L, with partition M-cubes centred in $M\mathbb{Z}^3/\ell\mathbb{Z}^3$, and assume $\ell/(LM)$ is odd and at least 13. A connection belongs to $\widetilde{\mathcal{G}}_k$ if $A = A_0 + A_1$, A_0 is real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, and on every concentric $11M$ enlargement of a partition cube,

$$A_0 = B_0 + d\lambda, \qquad B_0, \lambda \text{ real}, \qquad \|B_0\|_\infty < e_k^{-1+\varepsilon}, \qquad d\lambda(c, c') = \eta^{-1}(\lambda(c') - \lambda(c)).$$

The local gauges may be chosen independently.

Distances are periodic sup distances. For distinct chamber sites set $d_X(x, x') = \max(d(c, c'), \eta)$, and, on pairs with $0 < d_X < 1$, put

$$U_A(x, x') = \exp(ie_k \eta \tau A(c, c')), \qquad (\delta_{\alpha,A} v)(x, x') = \frac{U_A(x, x')v(x') - v(x)}{d_X(x, x')^\alpha}.$$

Thus distinct labels over one photon site are included when $k \geq 1$. For an operator T, define

$$N_\gamma(T) = \sup_z \sum_{z'} e^{\gamma d(z, z')} \|\mathbf{1}_{\Delta_z} T \mathbf{1}_{\Delta_{z'}}\|_{\infty \to \infty}.$$

The sup norm uses Euclidean spinor norms; for pair outputs the left indicator restricts the first point. At level zero, $N_\gamma^{\ker}$ replaces the localized operator norm by the supremum of the normalized chamber kernel on the two clusters.

Theorem A. There are $a' > 0$ and $0 < \gamma \le (6L)^{-1}$, depending only on ρ, L, α, with the following property. Set $C_V = 2\exp(1)(42 + 6b)$ and $q(p) = 54\pi C_V a' p/M$. Assume

$$M > 4L, \qquad q(1) \le \tfrac{1}{2}, \qquad 0 < e_k \le e_0, \qquad 6Le_0^{\varepsilon} \le 1, \qquad 4C_V a' e_0^{\varepsilon} \le 1. \tag{3}$$

For every $A \in \widetilde{\mathcal{G}}_k$, all the operators (2) are invertible. Their inverses $S_y(A)$ are analytic in the connection and satisfy

$$N_\gamma(S_y(A)) \le 32a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A} S_y(A)) \le 304a', \qquad w_k(y) = \begin{cases} 1, & k \ge 1, \\ (1+|y|)^{-1}, & k = 0. \end{cases}$$

At level zero the first bound also holds in N_γ^{ker}; there are no short pairs at that level. The constants are independent of the level, mesh, volume, permitted connection, reference mass $\mu \in (0, 1]$, and resolvent parameter in its stated range. ■□

We develop the chamber block arguments of [Cur, Sections 9 and 11]. The internal chamber graph has a positive gap on fields with zero cluster mean. Eliminating those components gives a four-component Schur symbol with the Dirac singularity at small momentum. Its Fourier bounds supply local regularity uniformly as the mesh tends to zero. Averaging penalties control the remaining block means. Local real gauges then reduce the connection to a small perturbation, and two local inverse identities give convergent right and left walk expansions.

The walk supports also localize the coarse covariance and determinant. We first attach parameters to the entire support of each walk, so that setting parameters to zero separates components. Exact resolvent identities pass from the fine inverse to the coarse inverse. Weighted Berezin integration then gives the block formula in theorem B. A resolvent integral expresses its determinant as a sum of local analytic terms. For the additional small-field difference estimate, long polymers supply the gain by decay; on short polymers a real gauge gives an enlarged complex disk, and Cauchy's formula gives the gain. The necessary nonwrapping geometry is proved in appendix A.

The separate free estimates below hold for arbitrary unitary period characters. The integration and transpose identities use the antiperiodic convention fixed here. Their spin matrices are

$$\mathsf{C} = i\gamma_2, \qquad \varkappa = \gamma_5 \mathsf{C} = \sigma_2 \otimes \sigma_2, \qquad \mathsf{C}^T = \mathsf{C}^{-1} = -\mathsf{C}, \quad \varkappa^T = \varkappa = \varkappa^{-1}.$$

Both matrices are real. Their displayed forms give the transpose and complex-conjugation identities used in the determinant calculation. The connection is an input to these estimates. For a connection obtained from Maxwell reconstruction, the local gauge and fluctuation domain estimates are supplied separately by [Max, Theorems A and B].

2. The free chamber inverse

The chamber estimates of [Cur, Section 9] give bounds which are uniform in the mesh, the volume, and the running mass. There are two sources of coercivity. The internal chamber graph controls the part of a spinor whose mean over a cluster vanishes. The averaging penalty controls the remaining four components. The first of these bounds also identifies the singular part of the free propagator. This permits a local regularity estimate with constants independent of the mesh, which is needed when the inverses on small tori are combined into an inverse on the full torus.

2.1. The cluster decomposition.

We first allow an arbitrary mesh $a > 0$, using the chamber stencil of section 1 with η replaced by a. Put $\widehat{d}_i(c, g) = -gn_i$ and

$$M_i(c, g) = \frac{w_i}{2}(\rho - \gamma \cdot \widehat{d}_i(c, g)).$$

The discussion applies on the covering lattice and on a torus of side $R \in a\mathbb{N}$, with any unitary character $f(c + Re_\mu, g) = -\exp(i\theta_\mu) f(c, g)$. Pairings on the torus are formed over a period cell. The

notation $d(c,c')$ denotes the sup distance modulo $R\mathbb{Z}^3$, or the covering sup distance when $R=\infty$. They use the measure

$$\int \mathrm{d}x\, v(x) = \frac{a^3}{48}\sum_c \sum_g v(c,g), \qquad \|f\|_2^2 = \int \mathrm{d}x\, |f(x)|_2^2.$$

On four-component spinors depending only on c, the measure is $a^3\sum_c$. With the cluster maps J, J^* of the introduction, set $P = JJ^*$ and $Q = 1-P$. The decomposition $f = J\xi + h$, with $\xi = J^*f$ and $h = Qf$, is orthogonal. In particular,

$$\|f\|_2^2 = \|\xi\|_2^2 + \|h\|_2^2. \tag{4}$$

The normalization by 48 makes J an isometry. Kernels on the chamber lattice are taken with respect to $\mathrm{d}x$, so the kernel of the identity is $48a^{-3}\mathbf{1}[x=x']1_4$.

Let A be a connection and write $u_A(c,c') = \exp(ie_a aA(c,c'))$ on a photon bond. The charge e_a in this notation is the charge at the level under consideration. The phases on the internal bonds are one. Put

$$R_i(A)f(x) = U_A(x,x^{(i)})f(x^{(i)}), \qquad \mathcal{L}_A = \sum_{i=0}^{3} w_i(1-R_i(A)).$$

The massless chamber operator is

$$D_A f(x) = -a^{-1}\sum_{i=0}^{3} M_i(x)\big(U_A(x,x^{(i)})f(x^{(i)}) - f(x)\big). \tag{5}$$

At level k, $a=\eta_k$ and $e_a = e_k$. The reference mass is added separately as $D_A+\mu$. For real A, the involutions $R_i(A)$ are unitary and self-adjoint, and the bond expansion gives

$$\mathrm{Re}(f, D_A f) = \frac{\rho}{2a}(f, \mathcal{L}_A f), \qquad D_A^* = \gamma_5 D_A \gamma_5, \qquad \gamma_5 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}. \tag{6}$$

Here γ_5 is Hermitian, $\gamma_5^2 = 1$, and it anticommutes with each γ_μ. Each unoriented bond contributes $a^3 w_i |U_A(x,x^{(i)})f(x^{(i)}) - f(x)|_2^2/48$ to the energy.

On one cluster let

$$\mathcal{L}_{\mathrm{int}} = \sum_{i=0}^{2} w_i(1-R_i), \qquad R_i v(g) = v(gs_i).$$

Its kernel consists of the constants: the three reflections s_0, s_1, s_2 generate O_{cub}. Denote its least positive eigenvalue by λ_{int} and set

$$g_{\mathrm{h}} = \frac{\rho\lambda_{\mathrm{int}}}{2}, \qquad c_* = \frac{\min(w_3, \lambda_{\mathrm{int}})}{18}, \qquad c_{\mathrm{lh}} = \sqrt{3}(1+\rho).$$

No numerical evaluation of λ_{int} is needed.

Proposition 2.1. For real A and $\xi = J^*f$,

$$(f, \mathcal{L}_A f) \geq \lambda_{\mathrm{int}}\|Qf\|_2^2, \qquad a^{-2}(f, \mathcal{L}_A f) \geq c_*\|\nabla_A \xi\|_2^2, \tag{7}$$

where $\nabla_{A,\mu}\xi(c) = a^{-1}(u_A(c, c+ae_\mu)\xi(c+ae_\mu) - \xi(c))$. The same inequalities hold on a finite set E of photon sites, with the energy restricted to bonds whose endpoints lie over E and the photon differences restricted to bonds in E. Moreover,

$$\|QD_A J\xi\|_2 \leq c_{\mathrm{lh}}\|\nabla_A\xi\|_2, \qquad |(\xi, J^*D_A Qf)| \leq c_{\mathrm{lh}}\|\nabla_A\xi\|_2\|f\|_2. \tag{8}$$

On $\operatorname{ran} Q$, for $t > -g_{\mathrm{h}}/a$,

$$\|(QD_AQ + t)^{-1}\|_{2\to 2} \leq (t + g_{\mathrm{h}}/a)^{-1}. \tag{9}$$

All constants are independent of a, R, the character, and the real connection. ∎

Proof. The internal bonds over c give $a^3(f(c,\cdot),\mathcal{L}_{\mathrm{int}}f(c,\cdot))$ and hence at least $\lambda_{\mathrm{int}}a^3\|Qf(c,\cdot)\|^2$. Summing over $c\in E$ proves the first bound in (7).

For the second bound, consider a photon bond $c'=c+av$. The eight labels with $ge_1=v$ correspond to the eight chamber bonds crossing it. Write $\mathcal{O}_v(c)$ for this octet. Since their endpoints form $\mathcal{O}_{-v}(c')$, the difference of the cluster means is the sum of the mean of the eight bond differences and two means of Qf:

$$u_A(c,c')\xi(c')-\xi(c)=\frac{1}{8}\sum_{x\in\mathcal{O}_v(c)}(U_A(x,x^{(3)})f(x^{(3)})-f(x))\\ -\frac{u_A(c,c')}{8}\sum_{x'\in\mathcal{O}_{-v}(c')}Qf(x')+\frac{1}{8}\sum_{x\in\mathcal{O}_v(c)}Qf(x).$$

The square is bounded by $3/8$ times the sum of the squared terms in these three octets. Distinct photon bonds use distinct chamber bonds; each octet of cluster remainders occurs at most once. Consequently, with $\mathcal{E}_3$ and $\mathcal{E}_{\mathrm{int}}$ denoting the two parts of the energy,

$$a^2\|\nabla_A\xi\|_{2,E}^2\le 18(w_3^{-1}\mathcal{E}_3+\|Qf\|_{2,E}^2)\le 18(w_3^{-1}\mathcal{E}_3+\lambda_{\mathrm{int}}^{-1}\mathcal{E}_{\mathrm{int}}).$$

This proves the local bound and therefore the global one.

The internal hops vanish on $J\xi$. Thus

$$(D_AJ\xi)(c,g)=-3(\rho-\gamma\cdot ge_1)\nabla_{A,ge_1}\xi(c).$$

Each of the six directions occurs eight times. The norm of the matrix in this formula is $3(1+\rho)$, and the two directed differences on a photon bond have equal norms. It follows that $\|D_AJ\xi\|_2^2\le 3(1+\rho)^2\|\nabla_A\xi\|_2^2$. Projection gives the first bound in (8); the second follows from (6), since J and Q commute with γ_5.

For $Qf=f$, (6) gives $\mathrm{Re}(f,(QD_AQ+t)f)\ge(t+g_{\mathrm{h}}/a)\|f\|_2^2$. The same lower bound holds for the adjoint. The operator has closed range and its adjoint has zero kernel, so it is onto, also on the infinite lattice. This proves (9). □

2.2. **The free symbol and its Schur complement.** Use lattice units for the moment. Let $\mathcal{D}_m(p)$ be the symbol of D_0+m, $0\le m\le 1$, on $V=\ell^2(O_{\mathrm{cub}};\mathbb{C}^4)$ with measure $48^{-1}\sum_g$:

$$(\mathcal{D}_m(p)v)(g)=mv(g)+\sum_{i=0}^{2}M_i(g)(v(g)-v(gs_i))+M_3(g)(v(g)-\exp(ip\cdot ge_1)v(gr_{e_1})).$$

Let $D_{\mathrm{ll}},D_{\mathrm{lh}},D_{\mathrm{hl}},D_{\mathrm{hh}}$ be its blocks in $V=\operatorname{ran}J\oplus\operatorname{ran}Q$. The mass is included in the diagonal blocks. Set

$$H_m(p)=D_{\mathrm{hh}}(p)^{-1},\qquad B_m(p)=D_{\mathrm{ll}}(p)-D_{\mathrm{lh}}(p)H_m(p)D_{\mathrm{hl}}(p),\qquad t(p)=\sum_\mu(1-\cos p_\mu).$$

The compression of the full operator is the Wilson operator:

$$D_{\mathrm{ll}}(p)=m+i\gamma\cdot\sin p+\rho t(p).$$

Its Schur complement also contains the contribution from the heavy part. In particular, adding a mass to the chamber operator changes H_m as well as D_{ll}.

Proposition 2.2. There are positive constants $\theta_0,c_{\mathrm{e}},\kappa_{\mathrm{strip}}$, depending on ρ only, such that H_m and B_m are holomorphic in a neighbourhood of $\{|\operatorname{Im}p_\mu|\le\theta_0\}$ and $2\pi\mathbb{Z}^3$-periodic. On the real cube

$[-\pi,\pi]^3$,

$$\|H_m(p)\| \le (g_{\mathrm{h}}+m)^{-1}, \tag{10}$$

$$\|D_{\mathrm{lh}}(p)\| + \|D_{\mathrm{hl}}(p)\| \le 2c_{\mathrm{lh}}(2t(p))^{1/2}, \tag{11}$$

$$B_m(p) = m + i\gamma\cdot\sin p + E_m(p), \qquad \|E_m(p)\| \le K_\rho t(p), \tag{12}$$

$$\mathrm{Re}(v, B_m(p)v) \ge (m+\rho c_* t(p))|v|_2^2, \tag{13}$$

$$|B_m(p)v|_2 \ge c_{\mathrm{e}}(|p|_2^2+m^2)^{1/2}|v|_2, \tag{14}$$

where $K_\rho = \rho + 2c_{\mathrm{lh}}^2/g_{\mathrm{h}}$. For every multi-index ν,

$$\|\partial_p^\nu B_m(p)^{-1}\| \le C_\nu(\rho)(|p|_2^2+m^2)^{-(1+|\nu|)/2}. \tag{15}$$

If $m>0$, the inverse exists also on $|\operatorname{Im} p_\mu| \le \kappa_{\mathrm{strip}} m$, where the same bound holds with $\operatorname{Re} p$ on the right, for $\operatorname{Re} p \in [-\pi,\pi]^3$. At $m=0$ the real derivative bound is asserted away from $p=0$. ■

Proof. Only the fourth hop depends on p. For $p = p' + ip''$,

$$\|\mathcal{D}_m(p) - \mathcal{D}_m(p')\| \le 3(1+\rho)(\exp(|p''|_\infty)-1).$$

Choose θ_0 so that the right side is $g_{\mathrm{h}}/2$ when $|p''|_\infty = \theta_0$. The heavy gap then gives $\operatorname{Re} D_{\mathrm{hh}}(p) \ge m + g_{\mathrm{h}}/2$ on this strip. Invertibility holds on a slightly larger strip, which proves holomorphy up to the closed strip. For real p, the off-diagonal bounds in proposition 2.1 give $\|D_{\mathrm{lh}}\|, \|D_{\mathrm{hl}}\| \le c_{\mathrm{lh}}(2t(p))^{1/2}$. This proves (12).

To retain the real-part bound after eliminating the heavy part, put $u = Jv - H_m D_{\mathrm{hl}} v$. Then $J^*u = v$, $Q\mathcal{D}_m u = 0$, and $\mathcal{D}_m u = JB_m v$. Therefore

$$\mathrm{Re}(v, B_m v) = \mathrm{Re}(u, \mathcal{D}_m u) = m\|u\|^2 + \frac{\rho}{2}(u, \mathcal{L}_0(p)u) \ge (m+\rho c_* t(p))|v|_2^2.$$

This proves (13).

The ellipticity estimate uses the Dirac part near zero and the real part away from zero. Put $t_0 = (4K_\rho^2)^{-1}$. One has $K_\rho \ge 48/\lambda_{\mathrm{int}} > 1$, since $\lambda_{\mathrm{int}} \le 12 + 24\sqrt{2} < 48$. Hence $t_0 \le 1/4$. If $t(p) \le t_0$, then $|\sin p|_2^2 \ge t(p)$ and $\|E_m(p)\| \le \frac{1}{2}t(p)^{1/2}$. Since

$$|(m + i\gamma\cdot\sin p)v|_2^2 = (m^2 + |\sin p|_2^2)|v|_2^2,$$

it follows that $|B_m(p)v|_2 \ge \frac{1}{2}(m^2+t(p))^{1/2}|v|_2$. On $[-\pi,\pi]^3$, $t(p) \ge 2|p|_2^2/\pi^2$. This proves (14) in this region. If $t(p) \ge t_0$, then (13) gives $|B_m(p)v|_2 \ge (m+\rho c_* t_0)|v|_2$. The inequality $(|p|_2^2+m^2)^{1/2} \le m + \pi\sqrt{3}$ closes the estimate with, for example,

$$c_{\mathrm{e}} = \min\left(0.22, \frac{\rho c_* t_0}{\pi\sqrt{3}}\right).$$

The strip bounds give uniform bounds on the derivatives of B_m by Cauchy's formula. Differentiating $B_m B_m^{-1} = 1$ and inducting on $|\nu|$ proves (15). For the complex strip, let K_s bound B_m on the fixed strip. On half that strip, $\|\partial_\mu B_m\| \le 2K_s/\theta_0$. Choose

$$\kappa_{\mathrm{strip}} = \min\left(\frac{\theta_0}{2}, \frac{c_{\mathrm{e}}\theta_0}{12K_s}\right).$$

Integration along the segment from p' to $p'+ip''$ shows that the change in B_m is at most $c_{\mathrm{e}}m/2$. The lower bound is thus at least $c_{\mathrm{e}}(|p'|_2^2+m^2)^{1/2}/2$. Differentiating the inverse identity once more gives the asserted complex derivative bounds. □

2.3. **Free kernels and coercivity.** Solving the heavy equation gives the inverse symbol

$$\mathcal{D}_m^{-1} = \begin{pmatrix} B_m^{-1} & -B_m^{-1} D_{\mathrm{lh}} H_m \\ -H_m D_{\mathrm{hl}} B_m^{-1} & H_m + H_m D_{\mathrm{hl}} B_m^{-1} D_{\mathrm{lh}} H_m \end{pmatrix}. \tag{16}$$

The first term is independent of the cluster labels after insertion by J and J^*. Every other term has a weaker singularity at $p = 0$: D_{lh} and D_{hl} vanish there, whereas H_m and all its derivatives are uniformly bounded.

Proposition 2.3. On the unit covering lattice let

$$G_m(c, g; c', g') = \int_{[-\pi,\pi]^3} \frac{\mathrm{d}p}{(2\pi)^3} \exp(ip \cdot (c - c'))[\mathcal{D}_m(p)^{-1}](g, g').$$

The bracket is the kernel on V relative to the label measure $48^{-1} \sum_g$, consistently with the chamber kernel convention. It is well-defined also at $m = 0$. It decomposes as $G_m(x, x') = G_m^{\mathrm{l}}(c - c') + G_m^{\mathrm{h}}(x, x')$, where

$$|G_m^{\mathrm{l}}(n)|_2 \le C_G(1 + |n|_\infty)^{-2}, \tag{17}$$

$$|G_m^{\mathrm{l}}(n + v) - G_m^{\mathrm{l}}(n)|_2 \le C_G(1 + |n|_\infty)^{-3}, \quad v \in \{\pm e_\mu\}, \tag{18}$$

$$|G_m^{\mathrm{h}}(x, x')|_2 \le C_G(1 + |c - c'|_\infty)^{-3}. \tag{19}$$

Consequently $|G_m(x, x')|_2 \le 2C_G(1 + |c - c'|_\infty)^{-2}$ and

$$|G_m(x^{(i)}, x') - G_m(x, x')|_2 \le 20 C_G(1 + |c - c'|_\infty)^{-3}. \tag{20}$$

For $m > 0$ these bounds have the additional factor $\exp(-\kappa_{\mathrm{strip}} m|c - c'|_\infty)$. The constants are uniform for $0 \le m \le 1$. ∎

Proof. The Fourier estimate used here can be proved directly. If a periodic symbol T satisfies $|\partial^\nu T(p)| \le A|p|_2^{-q-|\nu|}$, where $q < 3$ and $|\nu| \le N > 3 - q$, then its Fourier coefficients are bounded by $CA(1 + |n|_\infty)^{-(3-q)}$. For $n \neq 0$, put $r = |n|_\infty^{-1}$ and split the integral using a cutoff equal to one for $|p|_2 \le r$ and zero for $|p|_2 \ge 2r$. The inner integral is bounded by CAr^{3-q}. On the complement integrate N times in a coordinate with $|n_\mu| = |n|_\infty$. The differentiated integrand has integral at most CAr^{3-q-N}; the prefactor is r^N. Derivatives of the cutoff are supported in $r \le |p|_2 \le 2r$ and have the same bound. Boundary terms cancel by periodicity.

Apply this estimate with $(q, N) = (1, 3)$ to B_m^{-1}. Multiplication by $\exp(ip \cdot v) - 1$ lowers its singularity to order zero, for which $(q, N) = (0, 4)$ applies. In (16), the two off-diagonal terms have order zero and the term with two off-diagonal factors has order -1. The latter can also be bounded by the order-zero estimate. The remaining term H_m is holomorphic on the fixed strip, and its Fourier coefficients decay exponentially. The kernel normalization costs a factor 48 for operator-valued symbols on V; this is absorbed in C_G.

For $m > 0$, shift one coordinate of the contour by $i\kappa_{\mathrm{strip}} m \operatorname{sgn}(n_\mu)$, where $|n_\mu| = |n|_\infty$. The complex derivative bounds of proposition 2.2 give the same Fourier estimates on the shifted contour. This supplies the exponential factor. The fixed-strip bound for H_m is stronger than the required one.

An internal hop leaves c fixed, so its difference cancels G_m^{l}. An external hop replaces c by $c + ge_1$; use the light difference bound and the two heavy bounds at these points. The inequalities $1 + |n + v|_\infty \ge (1 + |n|_\infty)/2$ and $\exp(\kappa_{\mathrm{strip}} m) < \sqrt{3}$ give the factor 20 in (20). □

The Green kernel at mesh a, for $0 \le \mu a \le 1$, is $a^{-2} G_{\mu a}(x/a, x'/a)$. Thus

$$|G(x, x')|_2 \le 2C_G(a + |c - c'|_\infty)^{-2}, \qquad |G(x^{(i)}, x') - G(x, x')|_2 \le 20 C_G a(a + |c - c'|_\infty)^{-3}. \tag{21}$$

Fourier multiplication gives both inverse identities on finitely supported functions. The factor 48 in the identity kernel is consistent with the measure $\mathrm{d}x$ above.

For distinct sites put $d_X(x,\widetilde{x}) = \max(|c-\widetilde{c}|_\infty, a)$. The chamber graph has paths from x to $\widetilde{x}$ of at most $39 d_X(x,\widetilde{x})/a$ bonds, with photon sites in the coordinate box spanned by $c, \widetilde{c}$. Indeed, the generators s_0, s_1, s_2 are the simple reflections of the signed permutation group, with nine positive roots. Their Coxeter length is at most nine [Eti24, Theorem 22.14], so any two labels in one cluster are joined by at most nine internal bonds. A monotone photon path has at most $3|c-\widetilde{c}|_\infty/a$ steps. Before each step move to a label with the required external direction, and after the last step move to the final label. This uses at most $10 \cdot 3|c-\widetilde{c}|_\infty/a + 9 \le 39 d_X/a$ bonds. If d_X is less than half the distance to c', sum (21) along such a path. In the other case use the two size bounds. The resulting interpolation gives, for $0 \le \alpha \le 1$,

$$|G(x,x') - G(\widetilde{x},x')|_2 \le 6240 C_G d_X(x,\widetilde{x})^\alpha \left((a + |c-c'|_\infty)^{-2-\alpha} + (a+|\widetilde{c}-c'|_\infty)^{-2-\alpha} \right). \tag{22}$$

The following coercivity estimate is separate from the real-part bound. It is needed to recover mesh-independent control of the block remainder.

Proposition 2.4. There is $c_b = c_b(\rho) > 0$ such that

$$\|D_0 f\|_2^2 \ge c_b a^{-2} (f, \mathcal{L}_0 f). \tag{23}$$

For a cube E of side ℓ_E, let P_E be the mean over $E \times O_{\mathrm{cub}}$. Then

$$\|f - P_E f\|_{2,E}^2 \le C_P \ell_E^2 a^{-2} (f, \mathcal{L}_0 f)_E, \qquad C_P = \frac{1}{8c_*} + \frac{1}{\lambda_{\mathrm{int}}}. \tag{24}$$

■

Proof. The Schur estimate at zero mass implies $|B_0(p)v| \ge c_{\mathrm{l}} t(p)^{1/2} |v|$, with $c_{\mathrm{l}} = \rho c_*/(2K_\rho)$: use the Dirac part for $t(p) \le t_0$ and the real part for $t(p) \ge t_0$. Write $u = Jv + h$ in a fibre and $f = \mathcal{D}_0(p) u$. The heavy equation and then the light equation give

$$h = H_0(f_{\mathrm{h}} - D_{\mathrm{hl}} v), \qquad B_0 v = f_{\mathrm{l}} - D_{\mathrm{lh}} H_0 f_{\mathrm{h}}.$$

Since $t(p) \le 6$, these imply $(2t(p))^{1/2}|v| \le A_1 \|f\|$ and $\|h\| \le A_2 \|f\|$, where

$$A_1 = \frac{\sqrt{2}}{c_{\mathrm{l}}} \left(1 + \frac{6(1+\rho)}{g_{\mathrm{h}}} \right), \qquad A_2 = g_{\mathrm{h}}^{-1}(1 + c_{\mathrm{lh}} A_1).$$

The bond decomposition also gives

$$(u, \mathcal{L}_0(p) u) \le 6 t(p) |v|^2 + (30 + 24\sqrt{2}) \|h\|^2.$$

For the internal bonds the light part cancels, leaving at most $2(w_0 + w_1 + w_2)\|h\|^2$. In an external bond difference, split off the photon difference of v and the two heavy endpoint values, and bound the square by three times their squared norms. The octet count then gives $6t(p)|v|^2 + 3w_3\|h\|^2$. This proves the displayed upper bound. This yields (23) in each fibre with $c_b = (3A_1^2 + (30+24\sqrt{2})A_2^2)^{-1}$. Fourier decomposition gives the assertion for all characters on the torus and on the covering lattice, restoring the factor a^{-2}.

For (24), first apply the scalar lattice Poincaré bound $\|\xi - P_E\xi\|_{2,E}^2 \le \ell_E^2 \|\nabla_0 \xi\|_{2,E}^2/8$. In one dimension this follows by writing each difference as a sum of nearest-neighbour differences; the coefficient of a bond at position j is $j(n-j)/2 \le n^2/8$. In three dimensions use the three commuting line averages, whose successive remainders are orthogonal. Now $f - P_E f = J(\xi - P_E\xi) + Qf$. Its two parts are orthogonal, and the local bounds (7) give (24). The heavy term uses $\ell_E \ge a$. □

3. Local inverses and walk expansions

3.1. **Averaging penalties and the inverse at zero field.** Fix an odd $L \ge 5$ and put $a = \eta_k = L^{-k}$. Write e_k for the charge e_a at this mesh. The level data k, μ, R, e_k are independent, with $0 < e_k \le e_0$. Unit blocks have side one, and the next blocks have side L. In this and the following subsections a finite period R is a positive integer multiple of L. For $z \in \mathbb{Z}^3$ the unit block is

$\Delta_z = \{c \in a\mathbb{Z}^3 : |c - z|_\infty < 1/2\}$. For $z' \in L\mathbb{Z}^3$ the L-block is $E_{z'} = \{c \in a\mathbb{Z}^3 : |c - z'|_\infty < L/2\}$. These partitions are taken modulo $R\mathbb{Z}^3$ on the torus. Since L is odd, no fine site lies on a block boundary. Their means, including the cluster mean, are denoted Π_1 and Π_L. At nonzero connection the means use the gauge phases of section 1. With the constants and coefficients defined there, the blocked operators at zero connection are

$$F_y(0) = D_0 + \mu + \alpha_y \Pi_1 + \beta_y \Pi_L \quad (k \geq 1), \qquad F_y(0) = D_0 + \mu + b_L \Pi_L + i\gamma_5 y \quad (k = 0). \tag{25}$$

Here $0 \leq \mu \leq 1$. In the first formula $y \in \mathbb{R} \cup \{\infty\}$; in the second $y \in \mathbb{R}$. We use $-\infty = \infty$ when an adjoint identity reverses this parameter. The notation F_y denotes an operator in this section, not the curvature.

The penalty bounds follow by diagonalizing γ_5. On its eigenspace of sign ϵ, the two eigenvalues and their sum are

$$\frac{b_k i\epsilon y}{b_k + i\epsilon y}, \quad \frac{b_k^2 b_L}{(b_k + b_L + i\epsilon y)(b_k + i\epsilon y)}, \quad \frac{b_k(b_L + i\epsilon y)}{b_k + b_L + i\epsilon y}.$$

For the first eigenvalue v_1 and the sum v_2, direct calculation gives

$$|v_1|^2 = b_k \operatorname{Re} v_1, \qquad |v_2|^2 \leq b_k \operatorname{Re} v_2, \qquad \operatorname{Re} v_2 \geq \frac{b_k b_L}{b_k + b_L} \geq \frac{b_-}{L}.$$

Also $|\alpha_y| \leq b_k \leq b$ and $|\beta_y| \leq b_L$. On an L-block the decomposition $\mathrm{Pen} = \alpha_y(\Pi_1 - \Pi_L) + (\alpha_y + \beta_y)\Pi_L$ has orthogonal ranges. The scalar inequalities on the two spin eigenspaces therefore give

$$\operatorname{Re}(f, \operatorname{Pen} f)_E \geq \frac{b_-}{L} \|P_E f\|_{2,E}^2, \qquad \|\operatorname{Pen} f\|_{2,E}^2 \leq b \operatorname{Re}(f, \operatorname{Pen} f)_E. \tag{26}$$

The kernel on E is $\alpha_y \mathbf{1}[c, c' \text{ in the same unit block}] + \beta_y L^{-3}$, and its norm is at most $2b$.

Proposition 3.1. There are $C_I, C_5, \gamma_5' > 0$, depending on L, ρ and a fixed $\alpha \in (0, 1)$ only, such that $F_y(0)$ is invertible and

$$\|F_y(0)^{-1}\|_{2\to 2} \leq L C_I \qquad (k \geq 1).$$

If f is supported over one unit block $\Delta_{z'}$ and $v = F_y(0)^{-1} f$, then for x over Δ_z and $0 < d_X(x, \widetilde{x}) \leq 1$,

$$|v(x)|_2 + d_X(x, \widetilde{x})^{-\alpha} |v(x) - v(\widetilde{x})|_2 \leq C_5 \exp(-\gamma_5' d(z, z')) \|f\|_\infty. \tag{27}$$

This includes pairs in one cluster. The constants are uniform in k, R, the character, μ, and y. ∎

Proof. First suppose $k \geq 1$. Put $f = F_y(0)u$ and $X = \operatorname{Re}(u, f)$. The real-part identity and (26) imply

$$\sum_E \|P_E u\|_{2,E}^2 \leq \frac{L}{b_-} X, \qquad \|\operatorname{Pen} u\|_2^2 \leq bX, \qquad \|\mu u\|_2^2 \leq X.$$

Therefore $\|D_0 u\|_2 \leq \|f\|_2 + (1 + \sqrt{b})X^{1/2}$. Use (23) for $D_0 u$ and (24) on each L-block. With $K_1 = \sqrt{L} + \sqrt{Lb}$ this gives

$$\|u\|_2^2 \leq L \left(\frac{1}{b_-} + \frac{2C_P K_1^2}{c_b} \right) \|u\|_2 \|f\|_2 + \frac{2C_P}{c_b} L^2 \|f\|_2^2.$$

Hence $\|u\|_2 \leq L C_I \|f\|_2$, where $C_I = b_-^{-1} + 2C_P K_1^2 / c_b + (2C_P/c_b)^{1/2}$. The identity $F_y(0)^* = \gamma_5 F_{-y}(0) \gamma_5$ gives the same lower bound for the adjoint. Thus $F_y(0)$ has closed, dense range and is onto, including on the infinite lattice.

To obtain decay, let ψ be a bounded real periodic function with $|\psi(c) - \psi(c')| \leq a_0 |c - c'|_1 / L$. The mass commutes with this weight. The external hopping term gives $\|\exp(\psi)[D_0, \exp(-\psi)]\|_{2\to 2} \leq 3(1 + \rho)\exp(1) a_0 / L$. The penalty kernel is supported in one L-block, has norm at most $2b$, and $|\psi(c) - \psi(c')| \leq 3a_0$ there. Its contribution is at most $6L^4 b \exp(1) a_0 / L$. Put

$$c_w = \exp(1)(3(1 + \rho) + 6L^4 b), \qquad a_0 = \min(1/3, (2C_I c_w)^{-1}).$$

Applying the L^2 inverse bound and absorbing the commutator gives

$$\| \exp(\psi) F_y(0)^{-1} f \|_2 \le 2LC_I \| \exp(\psi) f \|_2. \tag{28}$$

Take truncated multiples of the ℓ^1 distance, divided by L, to the source L-block. Letting the truncation increase proves an exponentially weighted L^2 bound.

The passage from L^2 to the sup norm uses the free kernel. On the covering lattice choose $c^\circ \in a\mathbb{Z}^3$ and $r \ge 8a$, and put

$$B = \{|c - c^\circ|_\infty < r\}, \quad B^+ = \{|c - c^\circ|_\infty < r + a\}, \quad B' = \{|c - c^\circ|_\infty \le r/4\}.$$

Suppose u is defined over B^+ and $(D_0 + \mu)u = h$ over B, with $0 \le \mu a \le 1$. Both evaluation points below lie over B', and C depends on the fixed $\alpha \in (0,1)$ and ρ. Let χ equal one on the concentric half cube and zero outside the cube, with Lipschitz constant $2/r$. Then

$$\chi u = G(\chi h) + G([D_0, \chi]u).$$

The commutator contains only the external hop. Its rows are supported where $|c - c^\circ|_\infty > r/2 - a$. It satisfies $\|[D_0, \chi]u\|_1 \le 45 r^{1/2} \|u\|_{2,B^+}$. For a point of the quarter cube, this support is at distance at least $r/4 - a \ge r/8$. The bounds (21) and (22) consequently give

$$\begin{aligned} |u(x)|_2 &\le C(r\|h\|_{\infty,B} + r^{-3/2}\|u\|_{2,B^+}), \\ |u(x) - u(\widetilde{x})|_2 &\le C \left(\frac{d_X(x,\widetilde{x})}{r} \right)^\alpha (r\|h\|_{\infty,B} + r^{-3/2}\|u\|_{2,B^+}). \end{aligned} \tag{29}$$

For the first term, sum $(a + |c - c'|_\infty)^{-2}$ over the cube; the result is $O(r)$. For the difference, the analogous sum with exponent $2 + \alpha$ is $O(r^{1-\alpha}/(1-\alpha))$. The cluster average in $\mathrm{d}x'$ contributes one to both sums. For the commutator term, its support separation gives the size bound; summing bond differences along the chamber path gives the difference bound. These estimates explain why the constants do not grow when $a \downarrow 0$.

For $u = F_y(0)^{-1} f$, use $h = f - \operatorname{Pen} u$ in (29), with $r = L/4$. The enlarged cube meets at most eight L-blocks. On each, the penalty kernel and Cauchy–Schwarz give $\| \operatorname{Pen} u \|_\infty \le 2bL^{3/2} \|u\|_{2,E}$. The weighted bound (28) therefore bounds both terms in (29) by $CL \exp(-a_0 d_1(x, E')) \|f\|_\infty$, where $d_1(x, E')$ is the ℓ^1 distance to the source L-block divided by L. This applies for $k \ge 2$. For $k = 1$, the pointwise L^2 bound costs only $\sqrt{48} L^{3/2}$; L is fixed. Larger Hölder separations are controlled by the two sup bounds. Finally, $d_1(x, E') \ge d(z, z')/L - 3$, and the source unit block lies in E'. This proves (27) with $\gamma_5' = a_0/L$. The argument uses only periodic moduli and so applies to every unitary character. □

The first level has a different resolvent parameter. On an L-block the real-part bound and Poincaré inequality imply

$$\operatorname{Re}(u, F_y(0)u) \ge \frac{c_r}{L} \|u\|_2^2, \qquad c_r = \min(b_-, \rho/(2C_P L)).$$

The term $i\gamma_5 y$ has zero real part. Conjugation by a weight with Lipschitz constant a_z changes the hopping and penalty terms by at most $\exp(1)(6 + 3b)a_z$. Choose

$$a_z \le \min((3L)^{-1}, c_r/(2\exp(1)L(6 + 3b))).$$

The weighted inverse then has L^2 norm at most $2L/c_r$. On the other hand its part other than $i\gamma_5 y$ has norm at most $60 + b$. For $|y| \ge 2(60 + b)$, the inverse norm is at most $2/|y|$. Combining the two ranges and converting to a kernel, with its factor 48, gives

$$|F_y(0)^{-1}(x, x')|_2 \le C_{00}(1 + |y|)^{-1} \exp(-a_z d(c, c')). \tag{30}$$

This factor in y is needed when the determinant is represented by an integral of resolvents.

3.2. **Small connections and local inverses.** Use the covariant block means, short quotients, and row norms of section 1. Replacing D_0 by D_A and Π_1, Π_L by their covariant versions in (25) gives $F_y(A)$. For real A these idempotents are orthogonal projections. Inserting the block partition between two factors and using the triangle inequality for d proves $N_\gamma(TT') \le N_\gamma(T)N_\gamma(T')$. Enlarge C_5, C_{00} to at least one, and choose

$$\gamma = \tfrac{1}{2}\min(\gamma_5', a_z), \qquad a' = \max(C_5, C_{00}) \sum_{n\in\mathbb{Z}^3} \exp(-\gamma|n|_\infty).$$

The preceding estimates imply

$$N_\gamma(F_y(0)^{-1}) \le a' w_k(y), \qquad N_\gamma(\delta_{\alpha,0} F_y(0)^{-1}) \le a'. \tag{31}$$

At $k = 0$ the first bound holds also in N_γ^{ker}.

Proposition 3.2. Let B be a complex connection with $e_k\|B\|_\infty \le \delta$, where $3L\delta \le 1$ and $2C_V a'\delta \le 1$, with $C_V = 2\exp(1)(42 + 6b)$. Then $F_y(B)$ is invertible and

$$N_\gamma(F_y(B)^{-1}) \le 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,B} F_y(B)^{-1}) \le 18a'. \tag{32}$$

At $k = 0$ the first bound holds also in N_γ^{ker}. ■

Proof. Write $V = F_y(B) - F_y(0)$. The internal hops, the mass, and the resolvent term at level zero cancel. For the external hop, $|u_B - 1| \le \eta_k\delta\exp(\eta_k\delta)$; its weighted row sum is at most $42\exp(\gamma)\exp(\delta)\delta$. The two averaging penalties contribute respectively $3b\delta\exp(3\delta)$ and $3b\delta\exp(\gamma(L-1))\exp(3L\delta)$. Indeed, the nested paths have total length at most $3/2$, and the path to an L-block centre has length at most $3L/2$. Summing their normalized block kernels removes the numbers of sites. The bounds $\gamma L \le \log 2$ and $3L\delta \le 1$ give $N_\gamma(V) \le C_V\delta$.

Thus $N_\gamma(VF_y(0)^{-1}) \le 1/2$. The Neumann series gives

$$F_y(B)^{-1} = F_y(0)^{-1}\sum_{n\ge0}(-VF_y(0)^{-1})^n = \sum_{n\ge0}(-F_y(0)^{-1}V)^n F_y(0)^{-1}.$$

It proves the first bound in (32). The first form gives $N_\gamma(\delta_{\alpha,0}F_y(B)^{-1}) \le 2a'$. For distinct photon sites in a short pair, $d_X^{-\alpha}|U_B - 1| \le 3\delta\exp(3\delta) \le 1/4$; for intracluster pairs the expression is zero. Evaluation at the second point of a short pair uses at most 27 neighbouring blocks, and costs at most $27\exp(\gamma)$. Its contribution to the covariant quotient is at most $16a'$, proving the second bound. At level zero, the second form of the Neumann series keeps the kernel norm in the last factor, so the kernel estimate follows as well. □

Return to the side $\ell = R$ and the domain $\widetilde{\mathcal{G}}_k$ of section 1. On the concentric $3M$ enlargement $W_\square$ of an M-cube,

$$A = B + d\lambda, \qquad e_k\|B\|_\infty < 2e_k^\varepsilon, \qquad e_k\|\operatorname{Im} A\|_\infty < e_k^\varepsilon. \tag{33}$$

This follows by restricting $B_0 + A_1$ and λ to that cube. The domain is open: a sufficiently small complex perturbation can be added to A_1. It is invariant under real gauge transformations.

For a lift $z_\square$ of the cube centre put $W_\square = \{|c - z_\square|_\infty < 3M/2\}$ and $W_\square^- = \{|c - z_\square|_\infty < 3M/2 - \eta_k\}$. Embed $W_\square$ in the positive chamber of the periodic torus of side $6M$, whose lattice is translated by $\eta_k(1,1,1)/2$. Its level-j block centres are $L^j\eta_k(1,1,1)/2 + L^j\eta_k\mathbb{Z}^3$ for $0 \le j \le k+1$. Translation by $L(1,1,1)/2$ identifies these blocks with the unshifted nested blocks, because L^{k+1-j} is odd. The zero-field and small-connection estimates therefore apply on this torus with the same constants. Fold its photon coordinates across the planes at multiples of $3M$. Adjacent sites fold either to adjacent sites in $W_\square$ or to one site; in the latter case assign phase one. Pull back A by this fold to obtain A^W. The fold preserves the supremum norm and commutes with d on scalar functions. It also preserves the nested blocks, since $3M/(L^j\eta_k)$ is odd for $j \le k+1$.

Let ext extend a function over $W_\square$ by zero in the seven other chambers, and let res restrict to its embedded copy. Labels are unchanged in these two maps. Define

$$S_\square(A) = \operatorname{res} F_y(A^W)^{-1}\operatorname{ext}.$$

For $A = B + d\lambda$ as in (33), $A^W = B^W + d\lambda^W$, with λ^W real. Gauge conjugation by $\exp(ie_k\lambda^W)$ reduces the inverse to the small connection B^W. If $6Le_0^\varepsilon \le 1$ and $4C_V a' e_0^\varepsilon \le 1$, proposition 3.2 applies to every $0 < e_k \le e_0$. Restriction, extension, and unitary gauge factors do not increase the localized norms. Thus

$$N_\gamma(S_\square(A)) \le 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A} S_\square(A)) \le 18a'. \tag{34}$$

The second operator here is formed on the local cube before extension to the original torus. It does not differentiate a discontinuous extension by zero.

The local inverse has both identities needed for patching. If f is defined over $W_\square$, then

$$F_y(A)S_\square(A)f = f \quad \text{over } W_\square^-. \tag{35}$$

If f is supported over $W_\square^-$, then $F_y(A)f$ is supported over $W_\square$ and

$$S_\square(A)F_y(A)f = f. \tag{36}$$

For (35), the rows at an interior site use only its internal neighbours, its external neighbour, and its L-block. These lie in $W_\square$, where the original and folded operators agree. For (36), the support margin ensures that applying F_y cannot leave $W_\square$. After embedding, applying the folded operator to the extension of f agrees with extending $F_y(A)f$. Apply the inverse and restrict. For an antiperiodic spinor on the original torus, use one lift and then its signed translates; the enlarged cubes do not overlap their translates. These arguments also show that $S_\square$ reads A only in $W_\square$ and is analytic in those connection variables.

3.3. **The parametrix and the walk expansion.** Choose the product cutoff $h_\square$ with centre $z_\square$ and one-dimensional factor

$$h_0(t) = \begin{cases} 1, & |t| \le 1/4, \\ \cos(\pi(2|t| - 1/2)/2), & 1/4 \le |t| \le 3/4, \\ 0, & |t| \ge 3/4. \end{cases}$$

On a lift of the cube set

$$h_\square(c) = \prod_{\mu=1}^{3} h_0((c_\mu - z_{\square,\mu})/M), \qquad h_\square(c,g) = h_\square(c).$$

Extend this with period R and set it to zero away from the periodic images of the lifted cube neighbourhood. The cutoffs satisfy $\sum_\square h_\square^2 = 1$, $\operatorname{supp} h_\square \subset \{|c - z_\square|_\infty < 3M/4\}$, and $|h_\square(c) - h_\square(c')| \le \pi|c - c'|_1/M$. Put $C_\square = -[F_y(A), h_\square]$. Internal hops commute with $h_\square$, and the penalty has range at most L. The phase bounds and the same normalized sums used in the perturbation estimate give

$$N_\gamma(C_\square) \le \pi C_V/M. \tag{37}$$

The kernel is $(h_\square(c) - h_\square(c'))F_y(A)(x,x')$. Its two photon sites lie within $3M/4 + L$ of $z_\square$. If $M > 4L$, both lie in $W_\square^-$; multiplication by $h_{\square'}$ can give a nonzero operator only when $d(z_\square, z_{\square'}) \le M$.

Using (35) and (36) gives

$$F_y h_\square S_\square h_\square = h_\square^2 - C_\square S_\square h_\square, \tag{38}$$

$$h_\square S_\square h_\square F_y = h_\square^2 + h_\square S_\square C_\square. \tag{39}$$

Let $S^{\mathrm{p}} = \sum_{\square} h_{\square} S_{\square} h_{\square}$ and $R^{\mathrm{p}} = \sum_{\square} C_{\square} S_{\square} h_{\square}$. At most 27 cubes contribute to one row of R^{p}. Consequently,

$$N_\gamma(R^{\mathrm{p}}) \le q_0 := \frac{54\pi C_V a'}{M}.$$

Choose $M \ge 108\pi C_V a'$. Equation (38) now gives a right inverse by a convergent series:

$$S_y(A) = F_y(A)^{-1} = S^{\mathrm{p}} \sum_{n\ge 0} (R^{\mathrm{p}})^n. \tag{40}$$

The finite-dimensional operator $F_y(A)$ is therefore invertible. The left identity gives the second form with $R^{\mathrm{p}\prime} = -\sum_{\square} h_{\square} S_{\square} C_{\square}$.

The terms in (40) have a local description. A walk of length n is a sequence $\omega = (\square_0, \dots, \square_n)$ with successive centres at distance at most M. Its term is

$$S_\omega = (h_0 S_0 h_0)(C_1 S_1 h_1) \cdots (C_n S_n h_n),$$

where the subscript i denotes the cube $\square_i$. Other sequences give zero because of the support of the commutator. Its connection variables lie in $\bigcup_{i=0}^{n} W_{\square_i}$.

Proposition 3.3. For $p \ge 1$ with $q(p) = 54\pi C_V a' p/M \le 1/2$,

$$\sup_z \sum_{z'} \exp(\gamma d(z,z')) \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} S_\omega \mathbf{1}_{\Delta_{z'}}\| \le 16 a' p w_k(y) q(p)^n, \tag{41}$$

$$\sup_z \sum_{z'} \exp(\gamma d(z,z')) \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} \delta_{\alpha,A} S_\omega \mathbf{1}_{\Delta_{z'}}\| \le 152 a' p q(p)^n. \tag{42}$$

The norms are sup operator norms. At $k = 0$ the first bound holds also with the supremum of the normalized kernel over the two clusters. In particular, $S_y(A)$ is analytic on $\widetilde{\mathcal{G}}_k$ and

$$N_\gamma(S_y(A)) \le 32 a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A} S_y(A)) \le 304 a'. \tag{43}$$

These constants are independent of the level, the side, the running mass in $[0,1]$, and the connection in the stated domain. ∎

Proof. The first factor in a walk has norm at most $2a' w_k(y)$. Every additional factor has norm at most $2\pi C_V a'/M$ by (34) and (37). For a fixed output block, at most eight initial cutoffs contribute; there are at most 27 choices at each further step. Apply submultiplicativity row by row and sum these $8 \cdot 27^n$ bounds. This proves (41).

For the covariant quotient, only the factor at the left end of the product is differentiated. The product identity is

$$\delta_{\alpha,A}(h_{\square} v)(x,x') = h_{\square}(c) \delta_{\alpha,A} v(x,x') + d_X^{-\alpha}(h_{\square}(c') - h_{\square}(c)) U_A(x,x') v(x').$$

The first contribution has localized norm at most $18a'$. The second has coefficient at most $3\pi \exp(1)/M$, and evaluation at the second point costs at most $27\exp(\gamma)$. With the stated choice of M this contribution is at most a'. Thus $N_\gamma(\delta_{\alpha,A} h_{\square} S_{\square}) \le 19a'$. For intracluster pairs the second term vanishes, since $c = c'$. There are again at most eight possible initial cubes, including when the second point of a short pair enters the cutoff support. The count $8 \cdot 27^n$ now gives (42).

At level zero keep the kernel norm in the rightmost local inverse; the other factors are controlled in operator norm. The same submultiplicative row estimate proves the kernel assertion. Summing the geometric bounds with $p = 1$ gives (43). For a fixed torus the sum of the operator norms over all walks is finite; its tail of lengths at least j is bounded by $4a'(\ell/M)^3 2^{-j}$. Each walk term is analytic in the finitely many connection variables it reads. The uniform tail bound therefore proves analyticity of the sum. The local bounds in (41) contain no factor depending on ℓ. □

Proof of Theorem A. Choose γ, a' as in (31). The constants in the free estimates depend only on ρ, L, α. The two charge conditions give (34), and $q(1) \le 1/2$ gives the convergent parametrix. Proposition 3.3 supplies the analytic inverse and both asserted row estimates, including the normalized kernel estimate at level zero. □

4. Localized propagators and coarse covariances

We use the local integration construction of [Cur, Section 11]. The walk expansion provides its localization. A parameter is attached to each M-cube traversed by a walk, together with its one-layer neighbourhood. Setting parameters to zero then separates the operator into components. The coarse covariance is obtained from this expansion by an exact resolvent identity. At parameter value one it agrees with the exact covariance; its local terms expand the determinant and the interaction. We work at mesh $\eta_k = L^{-k}$ on the torus of side ℓ; this is the side denoted by R in the local inverse construction. Throughout this section the mass satisfies $0 < \mu \le 1$, the range of the renormalization data. Transposition is with respect to the bilinear lattice pairings: for $O : E \to E'$, $\langle v, Ou\rangle_{E'} = \langle O^T v, u\rangle_E$. These pairings use the counting, chamber, or coarse lattice measure of the respective space. No complex conjugation enters this operation.

4.1. Weakened walks and polymer propagators.

The resolvent parameter ranges over $Y_0 = \mathbb{R}$ at level zero and $Y_k = \mathbb{R} \cup \{\infty\}$ at $k \ge 1$, with $-\infty = \infty$ as in the preceding resolvent construction.

Let π_k be the partition into M-cubes. For integers $j \ge 0$, write $\square^{\sim j}$ for the union of partition cubes whose centres are within periodic sup distance jM of the centre of $\square$. When this enlargement does not wrap, it has side $(2j+1)M$. Let $\mathcal{P}_k$ be the set of nonempty unions of these cubes connected through faces. For $X \in \mathcal{P}_k$ write $|X|_M$ for its number of cubes. With the net and length conventions of appendix A, set

$$M d_M(X) = \inf\{\text{length}(\mathcal{N}) : \mathcal{N} \text{ is a net meeting every closed cube of } X\}.$$

For a face-connected union of cubes, joining successive cube centres gives $d_M(X) \le |X|_M - 1$.

For a walk $\omega = (\square_0, \dots, \square_n)$ from section 3.3, put

$$X_\omega = \bigcup_{i=0}^{n} \square_i^{\sim 1}, \qquad s_\omega = \prod_{\square \subset X_\omega} s_\square, \qquad |s_\square| \le M^{\alpha_0}, \quad \alpha_0 = 1/54.$$

The union X_ω is a polymer and

$$|X_\omega|_M \le 27(n+1), \qquad d_M(X_\omega) \le 27(n+1). \tag{44}$$

Indeed, each one-layer neighbourhood contains 27 cubes. Successive centres differ by at most M in each coordinate, so their neighbourhoods are joined through faces. The preceding bound on d_M applies. Every walk, including a walk of length zero, carries the parameters of all cubes in X_ω.

The left parametrix gives terms

$$S'_{y,\omega} = (h_n S_n)(-C_n h_{n-1} S_{n-1}) \cdots (-C_1 h_0 S_0) h_0.$$

Here, as for $S_{y,\omega}$, S_i is the local inverse on $\square_i$ and $C_i = -[F_y, h_i]$. Define

$$S_y(s, A) = \sum_\omega s_\omega S_{y,\omega}(A), \qquad S_y(X, A) = \sum_{X_\omega = X} S_{y,\omega}(A), \tag{45}$$
$$S'_y(s, A) = \sum_\omega s_\omega S'_{y,\omega}(A), \qquad S'_y(X, A) = \sum_{X_\omega = X} S'_{y,\omega}(A).$$

Their averages are denoted by S^{sy}_y. The superscript # below means any of these three choices. Put $\kappa' = \kappa_{\mathrm{w}} + \kappa_0$, where $\kappa_0 > 0$ and $\kappa_{\mathrm{w}} = \kappa + \kappa_0 + 3$, with $\kappa > 0$, are fixed polymer decay rates. Choose M large enough that

$$q(M^{27\alpha_0}) \le \tfrac{1}{2}, \qquad q(e^{27\kappa'}) \le \tfrac{1}{2}, \qquad q(p) = 54\pi C_V a' p/M. \tag{46}$$

Proposition 4.1. For $A \in \widetilde{\mathcal{G}}_k$, $y \in Y_k$, and $s \in \prod_{\pi_k}\{|s_\square| \le M^{\alpha_0}\}$, the series (45) converge absolutely in N_γ, uniformly in (s, A), and

$$N_\gamma(S_y^\#(s,A)) \le 32a'M^{27\alpha_0}w_k(y), \tag{47}$$

$$N_\gamma(\delta_{\alpha,A}S_y^\#(s,A)) \le 304a'M^{27\alpha_0},$$

$$\|\mathbf{1}_{\Delta_z}S_y^\#(X,A)\mathbf{1}_{\Delta_{z'}}\| \le 32a'e^{27\kappa'}w_k(y)e^{-\gamma d(z,z')-\kappa' d_M(X)}. \tag{48}$$

The norm in the last line is the sup operator norm. At $k = 0$ it can also be replaced by the supremum of the normalized kernel over the two clusters, with $w_0(y) = (1 + |y|)^{-1}$. At parameter value one,

$$S_y^\#(1,A) = S_y(A) = \sum_{X\in\mathcal{P}_k} S_y^\#(X,A). \tag{49}$$

The weakened operators are continuous on the closed parameter polydisk, jointly analytic in its interior and in A, and analytic in A for each fixed parameter value, including boundary values. For finite real y the families are continuous in y.

The operator $S_y^\#(X,A)$ is supported on X in both arguments and depends on A only on bonds with endpoints in X. If parameters vanish off a union Y of M-cubes, then each component of Y is preserved by $S_y^\#(s,A)$. On such a component the operator depends only on its parameters and its connection. Its row at $x = (c, g)$ vanishes unless all M-cubes meeting $K_c = \{c' : |c' - c|_\infty \le 2L\}$ have active parameters; the corresponding assertion holds for columns. ∎

Proof. By (44), $|s_\omega| \le p_s^{n+1}$ with $p_s = M^{27\alpha_0}$. Apply the walk row estimate with this weight. The length-n contribution is bounded by $16a'p_sw_k(y)2^{-n}$; its Hölder row estimate is bounded by $152a'p_s2^{-n}$. Summing these bounds proves the first two estimates. For a fixed polymer use

$$\mathbf{1}[X_\omega = X] \le e^{-\kappa' d_M(X)}p_X^{n+1}, \qquad p_X = e^{27\kappa'}.$$

The same row estimate with p_X proves (48). The left walk estimates follow from the left parametrix by the same weighted product estimate; taking the average preserves all bounds.

On a fixed torus, summing the row bounds over its finitely many unit blocks also bounds the sum of the N_γ norms of the individual terms. The tails decay geometrically, uniformly in (s, A). The series may therefore be regrouped by X_ω. At $s = 1$ they are the right and left inverse expansions, so both equal $S_y(A)$. This proves (49). Each finite sum is analytic in its connection variables and polynomial in s. The uniform tail estimates give the stated analyticity and continuity. The local inverse is continuous at every real resolvent parameter. Its walk terms inherit this property, and their tails are uniform in y since $w_k(y) \le 1$. The sums are therefore continuous in real y.

A walk term is a product of local factors on the neighbourhoods $\square_i^{\sim 1}$. Its row and column cutoffs, its commutators, and all connection variables consequently lie in X_ω. If the first row is nonzero, its photon site satisfies $|c - z_{\square_0}|_\infty < 3M/4$. Since $3M/4 + 2L < 3M/2$, every M-cube meeting K_c is in $\square_0^{\sim 1}$, and its parameter occurs in s_ω. The last cutoff gives the column assertion; the order is reversed for a left walk. Finally, a surviving walk has every cube of X_ω in Y. Since X_ω is face-connected, it lies in a single component of Y. Its local factors use only the connection in that component. This proves all the locality statements. □

The lattice symmetries used in this section have coordinate sign changes and translations in $M\mathbb{Z}^3$. Their spin lifts lie in the group generated by $\gamma_1, \gamma_2, \gamma_3, \gamma_5$; they act on (c, g) by $(rc + a, rg)$, where $S\gamma_\mu S^{-1} = r_{\mu\mu}\gamma_\mu$ determines r. Write $\widetilde{g} = (S, a)$ for such a symmetry. The step group restricts a to $LM\mathbb{Z}^3$. A lift commuting with γ_5 preserves y; a lift anticommuting with it sends y to $-y$ at nonneutral values. Gauge covariance follows term by term: the conjugating phases of adjacent factors cancel. A lattice symmetry permutes cubes and walks, and hence acts on the parameters by $(\widetilde{g}s)_\square = s_{\widetilde{g}^{-1}\square}$. Transposition reverses the order of the local factors and interchanges the right and

left expansions. More precisely, with $\mathsf{C} = i\gamma_2$ one has $F_y(A)^T = \mathsf{C}^{-1}F_{-y}(-A)\mathsf{C}$. Therefore

$$S_y(s,A)^T = \mathsf{C}^{-1}S'_{-y}(s,-A)\mathsf{C}. \tag{50}$$

Thus at the neutral values $y = 0, \infty$ for $k \geq 1$, and $y = 0$ for $k = 0$, the symmetrized family has the same transpose covariance as the exact inverse. This is the reason for using it in the coarse covariance and determinant.

4.2. **The fine and coarse resolvents.** For $k \geq 1$ let

$$\mathcal{Q}_k = Q_k(A)J^*, \qquad \mathcal{Q}_k^\sharp = JQ_k^T(-A), \qquad P_L = Q^T(-A)Q(A).$$

Here Q_k averages the fine lattice onto the unit lattice, and Q averages the unit lattice onto the lattice of spacing L. The notation $\sharp$ denotes the displayed algebraic transpose with reversed connection; for complex A it is not a Hermitian adjoint. Here $B(z') = E_{z'}$ is the L-block from the preceding section. To specify the phases, if $c \in \Delta_z$ and $z \in B(z')$, set

$$\varphi_A(z,c) = e^{ie_k\eta_k(\tau_k A)(z,c)}, \qquad U_A(z',z) = e^{ie_k\eta_k(\tau A)(z',z)}.$$

The path average τA is the arithmetic mean of the oriented bond sums in the six coordinate orders, and $\tau_k A$ is their sum along the nested block centres. Then

$$\begin{aligned}(\mathcal{Q}_k f)(z) &= \eta_k^3 \sum_{c\in\Delta_z} \varphi_A(z,c)\frac{1}{48}\sum_g f(c,g),\\ (\mathcal{Q}_k^\sharp v)(c,g) &= \varphi_A(z,c)^{-1}v(z),\\ (P_L v)(z) &= U_A(z',z)^{-1}L^{-3}\sum_{z''\in B(z')} U_A(z',z'')v(z'').\end{aligned}$$

The nesting of the odd blocks and cancellation of the phases give

$$\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1, \quad P_L^2 = P_L, \quad \mathcal{Q}_k^\sharp\mathcal{Q}_k = \Pi_1(A), \quad \mathcal{Q}_k^\sharp P_L\mathcal{Q}_k = \Pi_L(A). \tag{51}$$

All these operators commute with constant spin matrices.

Write $S_k = S_\infty(A)$ and define maps on the unit spinors by

$$\begin{aligned}H_k &= b_k S_k\mathcal{Q}_k^\sharp, & \mathfrak{H}_k &= b_k\mathcal{Q}_k S_k,\\ D_k &= b_k - b_k^2\mathcal{Q}_k S_k\mathcal{Q}_k^\sharp, & M_k &= b_k - D_k.\end{aligned} \tag{52}$$

For real y put

$$\mathcal{B}_{k,y} = (b_k + i\gamma_5 y)^{-1}(1-P_L) + (b_k + b_L + i\gamma_5 y)^{-1}P_L.$$

These 4×4 inverses exist since $(t+i\gamma_5 y)^{-1} = (t - i\gamma_5 y)/(t^2+y^2)$ for $t > 0$. The idempotent decomposition and (51) give

$$\mathcal{B}_{k,y}^{-1} = b_k + b_L P_L + i\gamma_5 y, \qquad F_y = F_\infty - b_k^2\mathcal{Q}_k^\sharp\mathcal{B}_{k,y}\mathcal{Q}_k. \tag{53}$$

Proposition 4.2. For $k \geq 1$, $A \in \widetilde{\mathcal{G}}_k$, and real y, the operator $D_k + b_L P_L + i\gamma_5 y$ is invertible and its inverse is

$$\Gamma_{k,y} = \mathcal{B}_{k,y} + b_k^2\mathcal{B}_{k,y}\mathcal{Q}_k S_y\mathcal{Q}_k^\sharp\mathcal{B}_{k,y}. \tag{54}$$

Moreover,

$$S_y = S_k + H_k\Gamma_{k,y}\mathfrak{H}_k. \tag{55}$$

In particular, $\Gamma_k = \Gamma_{k,0}$ exists and $\delta Z_k(A) = \det(D_k + b_L P_L) \neq 0$. ∎

Proof. Set $Z = b_k^2 \mathcal{Q}_k^\sharp \mathcal{B}_{k,y} \mathcal{Q}_k$. The two fine inverses exist by the preceding section, and (53) gives both identities

$$S_y - S_k = S_k Z S_y = S_y Z S_k.$$

Since $D_k + b_L P_L + i\gamma_5 y = \mathcal{B}_{k,y}^{-1} - b_k^2 \mathcal{Q}_k S_k \mathcal{Q}_k^\sharp$, multiplication by the right side of (54) gives the identity: the additional term is $b_k^2 \mathcal{Q}_k (S_y - S_k - S_k Z S_y) \mathcal{Q}_k^\sharp \mathcal{B}_{k,y} = 0$. The product in the other order uses $S_y - S_k - S_y Z S_k = 0$. This proves the inverse formula without assuming that the coarse operator is invertible. Substituting the formula in $H_k \Gamma_{k,y} \mathfrak{H}_k$ gives

$$S_k Z S_k + S_k Z S_y Z S_k = S_k Z (S_k + S_y Z S_k) = S_k Z S_y = S_y - S_k.$$

At $y = 0$ the coarse operator is an invertible map on the finite dimensional unit spinor space, so its determinant is nonzero. □

At level zero this algebra is replaced by a single chamber integration. The operators b_0, S_∞, $\mathcal{B}_{0,y}$, and M_0 are not defined. Instead,

$$F_0 = D_A + \mu + b_L J P_L J^*, \qquad \Gamma_0 = F_0^{-1}, \qquad \delta Z_0(A) = \det F_0 \neq 0. \tag{56}$$

Here Γ_0 acts on 192 components per photon site, whereas Γ_k for $k \geq 1$ acts on four components per unit site.

4.3. **Localized coarse covariances.** The weakened families for $k \geq 1$ are

$$\begin{aligned} H_k(s, A) &= b_k S_\infty(s, A) \mathcal{Q}_k^\sharp, \\ \Gamma_k(s, A) &= \mathcal{B}_{k,0} + b_k^2 \mathcal{B}_{k,0} \mathcal{Q}_k S_0^{\mathrm{sy}}(s, A) \mathcal{Q}_k^\sharp \mathcal{B}_{k,0}, \\ M_k(X, A) &= b_k^2 \mathcal{Q}_k S_\infty^{\mathrm{sy}}(X, A) \mathcal{Q}_k^\sharp. \end{aligned} \tag{57}$$

The unprimed polymer propagators $S_{k,y}(X, A) = S_y(X, A)$ are used in the determinant integral. At level zero let $\mathrm{Cub}(c)$ be the M-cubes having a lattice site in common with K_c, and set

$$\begin{aligned} (H_0(s, A)u)(c, g) &= \left(\prod_{\square \in \mathrm{Cub}(c)} s_\square \right) u(c, g), \\ \Gamma_0(s, A) &= S_0^{\mathrm{sy}}(s, A), \qquad S_{0,y}(X, A) = S_y(X, A). \end{aligned} \tag{58}$$

At $s = 1$ these are the exact operators; also $\sum_X M_k(X, A) = M_k(A)$. The exponential formulas for the averaging phases and proposition 4.1 give analyticity on the same domains.

For use in the estimates, let $\mathfrak{i}(A) = e_k \| \operatorname{Im} A \|_\infty$. The small-connection domain and the charge bound give $\mathfrak{i}(A) \leq 1/(6L)$; reduce γ, if necessary, so that $\gamma L \leq 1/6$. The path lengths imply

$$|\varphi_A(z, c)^{\pm 1}| \leq e^{3\mathfrak{i}(A)/2}, \qquad \|\mathcal{Q}_k\|_{\infty\to\infty}, \|\mathcal{Q}_k^\sharp\|_{\infty\to\infty} \leq e^{3\mathfrak{i}(A)/2}. \tag{59}$$

The sup norms here are formed with the Euclidean spinor norm. Each unit block has chamber measure one. In particular, for a fine operator O,

$$|(\mathcal{Q}_k O \mathcal{Q}_k^\sharp)(z, z')|_2 \leq \exp(1) \, \|\mathbf{1}_{\Delta_z} O \mathbf{1}_{\Delta_{z'}}\|. \tag{60}$$

Indeed, apply the two averaging bounds to a unit spinor supported at z'; their product is $e^{3\mathfrak{i}(A)} \leq \exp(1)$.

Proposition 4.3. The families (57) satisfy

$$\|H_k(s, A)\|_{\infty\to\infty} \leq 32 \exp(1) b_k a' M^{27\alpha_0}, \tag{61}$$

$$\begin{aligned} \|\delta_{\alpha,A} H_k(s, A)\|_{\infty\to\infty} &\leq 304 \exp(1) b_k a' M^{27\alpha_0}, \\ |\Gamma_k(s, A)(z, z')|_2 &\leq C_\Gamma M^{27\alpha_0} e^{-\gamma d(z,z')}, \end{aligned} \tag{62}$$

$$|M_k(X, A)(z, z')|_2 \leq 32 \exp(1) b_k^2 a' e^{27\kappa'} e^{-\gamma d(z,z') - \kappa' d_M(X)},$$

where

$$C_\Gamma = \frac{2(1+2\exp(1))}{b_-} + 128\exp(1)(1+2\exp(1))^2 a'.$$

The Hölder estimate is on short pairs, including the distinct sites of one cluster. The families $H_k(s,A)$ and $\Gamma_k(s,A)$ preserve every component of the active M-cubes and depend there only on its parameters and connection. The row of $H_k(s,A)$ at (c,g) vanishes unless all cubes meeting K_c are active. The polymer $M_k(X,A)$ is supported on the unit sites of X and depends only on the connection in X.

At level zero the normalized kernel of $\Gamma_0(s,A)$ is bounded by $32a'M^{27\alpha_0}e^{-\gamma d(c,c')}$, and (48) bounds $S_{0,y}(X,A)$ with its factor $(1+|y|)^{-1}$. The multiplier $H_0(s,A)$ has sup norm at most $M^{8\alpha_0}$ and equals the identity at $s=1$. Its row at (c,g) vanishes unless every cube meeting K_c is active. There are no short pairs of distance less than one at this level. ■

Proof. The first two estimates follow by composing the weakened fine bounds with (59). The third and fourth use (60). For the covariance it remains to control the two factors $\mathcal{B}_{k,0}$. Its kernel is

$$\mathcal{B}_{k,0}(z,z') = b_k^{-1}\mathbf{1}[z=z']1_4 - \frac{b_L}{b_k(b_k+b_L)}P_L(z,z').$$

It vanishes unless z,z' belong to one L-block. Since $|P_L(z,z')|_2 \le \exp(1)L^{-3}$ on that block, its row and column sums are at most $(1+2\exp(1))/b_k$. In a nonzero term of $\mathcal{B}_{k,0}G\mathcal{B}_{k,0}$, where $G = \mathcal{Q}_k S_0^{\mathrm{sy}}(s,A)\mathcal{Q}_k^\sharp$, the two intermediate sites are within L of the exterior sites. Therefore

$$e^{-\gamma d(z_1,z_2)} \le e^{2\gamma L}e^{-\gamma d(z,z')} \le 4e^{-\gamma d(z,z')}.$$

The fine bound and (60) give $|G(z_1,z_2)|_2 \le 32\exp(1)a'M^{27\alpha_0}e^{-\gamma d(z_1,z_2)}$. Multiplying the row and column sums cancels the factor b_k^2 and gives $128\exp(1)(1+2\exp(1))^2a'M^{27\alpha_0}$. The local term $\mathcal{B}_{k,0}$ is bounded by $2(1+2\exp(1))b_-^{-1}e^{-\gamma d(z,z')}$, proving (62).

The odd block partitions are nested, and every unit block and every L-block belongs to a single M-cube. Moreover, the paths entering $\varphi_A(z,c)$ lie in Δ_z, and the paths entering $U_A(z',z)$ lie in the fine block of side L centred at z': each coordinate of a rectilinear path remains between those of its endpoints. Consequently the averaging maps and $\mathcal{B}_{k,0}$ preserve every union of M-cubes and depend only on its connection. Composing them with the component-local fine families proves the locality assertions. This applies also to the parameter-independent term $\mathcal{B}_{k,0}$ in $\Gamma_k(s,A)$. For the row of H_k, the terminal average does not change the first row index of $S_\infty(s,A)$, so the stronger vanishing statement is preserved. At level zero the assertions for Γ_0 follow from the fine kernel bound and locality with $\eta_0 = 1$.

For H_0, lift c and K_c to the lattice. In each coordinate K_c is an interval of $4L+1$ sites. Since $M \ge L^2$ and $L \ge 5$, this interval meets at most two consecutive M-blocks. The cubes meeting K_c are the product of these three sets of blocks. They number at most eight, form a face-connected set, and include the cube of c. Thus the defining product is a monomial of degree at most eight, bounded by $M^{8\alpha_0}$. If one of its cubes is inactive, its factor is zero. If all are active, their connectedness puts them in the active component containing c; hence changing parameters outside that component has no effect on the row or column at (c,g). The multiplier is scalar in spin and chamber labels and independent of the connection. These observations give its analyticity, gauge covariance, and component locality. □

These families have the gauge and lattice covariance of their exact counterparts. For Γ_k and M_k, transposition follows from (50) at a neutral value and the scalar averaging phases. The barred field map is $\bar{H}_k(s,A) = \mathsf{C}^{-1}H_k(s,-A)\mathsf{C}$. Its bounds follow from those for H_k because C is unitary and $\mathsf{C}\gamma_\mu\mathsf{C}^{-1} = -\gamma_\mu^T$. When component sup norms are used in the Grassmann substitution, the Euclidean operator bounds above acquire at most the factor two from $|v|_c \le |v|_2 \le 2|v|_c$.

5. Weighted Grassmann block integration

5.1. **Weighted Grassmann integration.** Let $\phi_i(t)$, $1 \le i \le p$, be independent Grassmann families on finite ordered index sets. If $\phi_i(I_i)$ is the ordered product over I_i, write

$$F = \sum_{I_1,\dots,I_p} c_F(I_1,\dots,I_p)\phi_1(I_1)\cdots\phi_p(I_p), \qquad \|F\|_{\mathsf{w}} = \sum_I |c_F(I)| \prod_i \mathsf{w}_i^{|I_i|}.$$

The positive measures $\nu_i(t)$ on the index sets enter the kernels, not this coefficient norm. Thus an antisymmetric kernel expansion

$$F = \sum_n \prod_i \frac{1}{n_i!} \int \mathrm{d}t\, K_n(t) \prod_i \prod_{j=1}^{n_i} \phi_i(t_j^i)$$

has norm

$$\|F\|_{\mathsf{w}} = \sum_n \prod_i \frac{\mathsf{w}_i^{n_i}}{n_i!} \int \mathrm{d}t\, |K_n(t)|. \tag{63}$$

For arbitrary kernels the right side is an upper bound. Products of ordered monomials vanish on repeated generators and otherwise multiply their weights, so

$$\|FG\|_{\mathsf{w}} \le \|F\|_{\mathsf{w}}\|G\|_{\mathsf{w}}, \qquad \|e^F - 1\|_{\mathsf{w}} \le e^{\|F\|_{\mathsf{w}}} - 1 \quad (F \text{ even}). \tag{64}$$

For an additional finite nonempty mode set $\mathcal{I}$ with positive weights $\nu(t)$, introduce $W(t), \bar W(t)$ and set

$$\langle \bar W, OW\rangle_\nu = \sum_{t,t'} \nu(t)\nu(t')\bar W(t)O(t,t')W(t'), \qquad \mathrm{Top} = \prod_{t\in\mathcal{I}} W(t)\bar W(t).$$

The kernel represents O with measure ν; its matrix entry is $O(t,t')\nu(t')$. Define $\int \mathrm{D}W$ as extraction of the coefficient of Top, multiplied by $\prod_t \nu(t)^{-1}$, and define

$$\int \mathrm{d}\mu_I(W) F = \int \mathrm{D}W\, e^{-\langle \bar W, W\rangle_\nu} F, \tag{65}$$
$$\int \mathrm{d}\mu_C(W) F = (\det C) \int \mathrm{D}W\, e^{-\langle \bar W, C^{-1}W\rangle_\nu} F \quad (C \text{ invertible}).$$

These are algebra-valued functionals when other Grassmann generators are present. The order of modes in Top is immaterial because its factors are even.

Proposition 5.1. Put $w_W = \max_t \nu(t)^{-1/2}$. Then

$$\left\| \int \mathrm{d}\mu_I(W) F \right\|_{\mathsf{w}} \le \|F\|_{(\mathsf{w}, w_W)}. \tag{66}$$

The integral factors for polynomials involving disjoint sets of modes. For an invertible covariance,

$$\int \mathrm{d}\mu_C(W) F(\bar W, W) = \int \mathrm{d}\mu_I(W) F(\bar W, CW), \tag{67}$$
$$\int \mathrm{d}\mu_C(W) W(t)\bar W(t') = C(t,t'),$$
$$\int \mathrm{D}W\, e^{-\langle \bar W, OW\rangle_\nu} = \det O \quad \text{for every } O. \tag{68}$$

Finally, consider a linear substitution $\phi_i \mapsto \sum_j O_{ij}\theta_j$, and put

$$\|O_{ij}\|_{1,\infty} = \sup_t \sum_s |O_{ij}(t,s)|\nu_j'(s).$$

If $\sum_j \|O_{ij}\|_{1,\infty}\mathsf{w}_j' \le \mathsf{w}_i$ for each i, then its image F' obeys $\|F'\|_{\mathsf{w}'} \le \|F\|_{\mathsf{w}}$. Only rows corresponding to generators occurring in F need satisfy this bound. ∎

Proof. The identity density factors as

$$e^{-\langle \bar{W}, W\rangle_\nu} = \prod_t (1 + \nu(t) W(t) \bar{W}(t)).$$

An ordered monomial $W(J)$ contributes to its integral only if J contains both generators of every mode it meets. In that case its integral is its reordering sign times $\prod_{(t,0)\in J} \nu(t)^{-1}$; otherwise it is zero. Its modulus is therefore at most $\prod_{u\in J} \nu(u)^{-1/2} \le w_W^{|J|}$. Apply this coefficientwise to obtain (66). For disjoint mode sets a nonzero integral requires each monomial separately to be paired. Their degrees in W are then even, so the signs from exchanging external generators cancel. The paired-mode formula gives factorization. It also shows that modes absent from the integrand may be removed.

A linear change $W \mapsto OW$ multiplies the coefficient of the top monomial by $\det O$, since the matrix of the change is $O(t,t')\nu(t')$. Applied to the identity density this proves (68), without an invertibility assumption. For invertible C, use $W \mapsto CW$, leaving $\bar{W}$ fixed, in the density with covariance C. This proves (67) and normalization. Its two-point function is

$$\sum_s C(t,s)\nu(s) \int \mathrm{d}\mu_I(W) W(s) \bar{W}(t') = \sum_s C(t,s)\nu(s) \frac{\mathbf{1}[s = t']}{\nu(t')} = C(t,t').$$

Thus it is the kernel, rather than the matrix entry.

Under the stated substitution, the norm of the image of $\phi_i(t)$ is at most

$$\sum_j \mathsf{w}'_j \sum_s |O_{ij}(t,s)| \nu'_j(s) \le \mathsf{w}_i$$

by hypothesis. Submultiplicativity bounds the image of each monomial by its original weight; summing its coefficients proves the last assertion, with the same proof on the subalgebra generated by the rows that occur. □

For linear combinations $v_1, \dots, v_{2r}$ of the $W, \bar{W}$, Wick's formula gives

$$\int \mathrm{d}\mu_I(W) v_1 \cdots v_{2r} = \mathrm{Pf}\left(\int \mathrm{d}\mu_I(W) v_i v_j \right)_{i,j}. \tag{69}$$

Odd products integrate to zero. To fix the signs, let ∂_u be the left Grassmann derivative. It satisfies $\partial_u(vG) = (\partial_u v)G - v\partial_u G$, and its Berezin integral is zero. Differentiating the factored Gaussian density gives

$$\int \mathrm{d}\mu_I(W) W(t) G = \nu(t)^{-1} \int \mathrm{d}\mu_I(W) \partial_{\bar{W}(t)} G,$$
$$\int \mathrm{d}\mu_I(W) \bar{W}(t) G = -\nu(t)^{-1} \int \mathrm{d}\mu_I(W) \partial_{W(t)} G.$$

Consequently, if $I(v) = \int \mathrm{d}\mu_I(W) v$,

$$I(v_1 \cdots v_q) = \sum_{j=2}^{q} (-1)^j I(v_1 v_j) I(v_2 \cdots \widehat{v}_j \cdots v_q).$$

This is the expansion of the Pfaffian along its first row and proves (69) by induction. Formula (67) gives the corresponding Wick rule for μ_C.

In the renormalization step the modes have weight one for $k \ge 1$. At $k = 0$ they are (c, g, β) with weight $1/48$. Thus the integration weight is

$$w_W = 1 \quad (k \ge 1), \qquad w_W = \sqrt{48} \quad (k = 0). \tag{70}$$

In particular the first-step identity covariance has two-point function $48\mathbf{1}[(c,g,\beta) = (c',g',\beta')]$. The pair index used for a Hölder field has measure $(\eta_k^3/48)^2$. In a substitution involving $\delta_{\alpha,A} H_k$, its scalar

component row sum is computed with the input measure, as in proposition 5.1. Internal-cluster differences are therefore controlled by the Hölder bound with the same field weight as other pairs.

5.2. **The Gaussian block identity.** For this identity use a common notation for the integration space: at $k \geq 1$ let $\mathcal{D}_k = D_k$ and $\mathcal{A}_k(A) = Q(A)$; at $k = 0$ let $\mathcal{D}_0 = D_A + \mu$ and $\mathcal{A}_0(A) = Q(A)J^*$. The integration measure is respectively counting measure or the chamber measure $1/48$. The target is the four-component spinor space on the lattice of spacing L, with measure L^3. Set

$$\mathcal{A}_k^\sharp(A) = \mathcal{A}_k(-A)^T, \qquad K_k = \mathcal{D}_k + b_L \mathcal{A}_k^\sharp \mathcal{A}_k, \qquad \Gamma_k = K_k^{-1}.$$

Thus $K_k = D_k + b_L P_L$ for $k \geq 1$ and $K_0 = F_0$. Define the quadratic operator on the coarse lattice by

$$D_{k+1}^0 = b_L - b_L^2 \mathcal{A}_k \Gamma_k \mathcal{A}_k^\sharp.$$

Let $\Phi, \bar{\Phi}$ be external coarse Grassmann fields and put

$$\Psi^c = b_L \Gamma_k \mathcal{A}_k^\sharp \Phi, \qquad \bar{\Psi}^c = b_L \Gamma_k^T \mathcal{A}_k(A)^T \bar{\Phi}.$$

Theorem B. Assume the hypotheses of theorem A and (46), with $\kappa_{\mathrm{w}} = \kappa + \kappa_0 + 3$ and $\kappa, \kappa_0 > 0$. Let $A \in \widetilde{\mathcal{G}}_k$ and $0 < \mu \leq 1$, with fermions antiperiodic in every coordinate. Then K_k is invertible. For every Grassmann polynomial F and external coarse fields $\Phi, \bar{\Phi}$, the weighted integral is

$$\int \mathrm{D}\Psi \, e^{-\langle \bar{\Psi}, \mathcal{D}_k \Psi \rangle - b_L \langle \bar{\Phi} - \mathcal{A}_k(-A)\bar{\Psi}, \Phi - \mathcal{A}_k(A)\Psi \rangle_L} F(\bar{\Psi}, \Psi)$$
$$= (\det K_k) e^{-\langle \bar{\Phi}, D_{k+1}^0 \Phi \rangle_L} \int \mathrm{d}\mu_I(W) F(\bar{\Psi}^c + \bar{W}, \Psi^c + \Gamma_k W). \quad (71)$$

The incoming measure is one per unit-site spin component for $k \geq 1$, and $1/48$ per chamber spin component for $k = 0$; the latter integration includes all 192 components per photon site. The target pairing uses measure L^3. The identity-Gaussian integration weights are respectively 1 and $\sqrt{48}$.

There are analytic functions $E_k^d(X, A)$, depending only on bonds meeting the polymer X, such that

$$\frac{\det K_k(A)}{\det K_k(0)} = \exp \sum_{X \in \mathcal{P}_k} (E_k^d(X, A) - E_k^d(X, 0)), \qquad |E_k^d(X, A)| \leq C_d M^3 e^{-(\kappa+1) d_M(X)}. \quad (72)$$

They are invariant under real gauge transformations, even in A, and satisfy $\overline{E_k^d(X, \overline{A})} = E_k^d(X, A)$. They are covariant under coordinate sign changes and translations in $LM\mathbb{Z}^3$. The constant C_d depends on the fixed $\rho, L, \alpha, M, \kappa, \kappa_0$, and is independent of the level, volume, connection in the stated domain and $\mu \in (0, 1]$. ■□

The block identity follows by completing the quadratic form with the weighted transposes. Its determinant is then localized by the resolvent integral in section 6.1; the polymer estimates permit that integral to be taken term by term.

For $\Psi = \Psi^c + W'$ and $\bar{\Psi} = \bar{\Psi}^c + \bar{W}'$, completion of the square is the exact identity

$$\langle \bar{\Psi}, \mathcal{D}_k \Psi \rangle + b_L \langle \bar{\Phi} - \mathcal{A}_k(-A)\bar{\Psi}, \Phi - \mathcal{A}_k(A)\Psi \rangle_L$$
$$= \langle \bar{\Phi}, D_{k+1}^0 \Phi \rangle_L + \langle \bar{W}', K_k W' \rangle. \quad (73)$$

Indeed, write $s = b_L \mathcal{A}_k^\sharp \Phi$ and $\bar{s} = b_L \mathcal{A}_k(A)^T \bar{\Phi}$. The left side is

$$\langle \bar{\Psi}, K_k \Psi \rangle - \langle \bar{\Psi}, s \rangle - \langle \bar{s}, \Psi \rangle + b_L \langle \bar{\Phi}, \Phi \rangle_L.$$

Substitute the means $\Gamma_k s, \Gamma_k^T \bar{s}$. Both cross terms cancel, and the remaining constant is $b_L \langle \bar{\Phi}, \Phi \rangle_L - \langle \bar{s}, \Gamma_k s \rangle = \langle \bar{\Phi}, D_{k+1}^0 \Phi \rangle_L$. All factors remain in their displayed order; no conjugation of Grassmann fields is involved.

Translation by external odd fields preserves the top coefficient in the integration variables. Applying this to (73) and then using (67) proves (71), with $\delta Z_k(A) = \det K_k(A)$. At the first step this is a single integration over all chamber components. The determinant and the integration weight are those of (56) and (70); the heavy components are included in both.

6. The local determinant

6.1. **An integral for the determinant.** The determinant in (71) has a local logarithm obtained by inserting the polymer propagators into a resolvent integral. We first derive the integral at real connections. Put $n_k = \ell^3$ and

$$c_k^d = 4\big((1-L^{-3})\log b_k + L^{-3}\log(b_k+b_L)\big) \quad (k\ge 1), \qquad c_0^d = 0.$$

For real $A \in \widetilde{\mathcal{G}}_k$ and $0 < \mu \le 1$,

$$\delta Z_k(A) = \exp\left\{c_k^d n_k - i b_k^2 \int_0^\infty \mathrm{d}y\ \mathrm{Tr}\big[\gamma_5 \mathcal{B}_{k,y} \mathcal{Q}_k S_y \mathcal{Q}_k^\sharp \mathcal{B}_{k,y}\big]\right\} \quad (k\ge 1), \tag{74}$$

$$\delta Z_0(A) = \exp\left\{\int_1^\infty \frac{\mathrm{d}y}{y}\,\mathrm{Tr}[F_0 S_y] - i\int_0^1 \mathrm{d}y\ \mathrm{Tr}[\gamma_5 S_y]\right\}. \tag{75}$$

The trace in the first line is on the unit spinor space. At level zero it is the normalized chamber trace $\mathrm{Tr}\, O = 48^{-1}\sum_{c,g} \mathrm{tr}\, O((c,g),(c,g))$.

To prove these identities, write $K = K_k$ as in the preceding subsection. At real A, the fine adjoint identity $D_A^* = \gamma_5 D_A \gamma_5$, the reality of μ, b_k, b_L, and the orthogonal averaging projections give $K^* = \gamma_5 K \gamma_5$. Hence $K\gamma_5$ is Hermitian and invertible. Since $\det \gamma_5 = 1$ on each four-component spinor, its determinant is $\det K$. Let λ_j be its real nonzero eigenvalues, counted with multiplicity. There are $4n_k$ at positive levels and $192 n_0$ at level zero. Then

$$\mathrm{Tr}\big[\gamma_5 (K+i\gamma_5 y)^{-1}\big] = \sum_j (\lambda_j + iy)^{-1}, \qquad \mathrm{Tr}\big[K(K+i\gamma_5 y)^{-1}\big] = \sum_j \lambda_j(\lambda_j+iy)^{-1}.$$

For a real $\lambda \ne 0$, choose along $y \ge 0$ the logarithm with value $\log|\lambda| + i\pi \mathbf{1}[\lambda < 0]$ at zero. Its derivative is $i/(\lambda+iy)$ and $\mathrm{Log}(\lambda+iy) - \log y \to i\pi/2$. Integration gives

$$-i\int_0^Y \frac{\mathrm{d}y}{\lambda+iy} = \mathrm{Log}\,\lambda - \mathrm{Log}(\lambda+iY),$$

$$\int_1^\infty \frac{\mathrm{d}y}{y}\frac{\lambda}{\lambda+iy} - i\int_0^1 \frac{\mathrm{d}y}{\lambda+iy} = \mathrm{Log}\,\lambda - i\pi/2.$$

Summing the second identity proves (75): its constant phase is $-96\pi i n_0 \in 2\pi i\mathbb{Z}$.

For $k \ge 1$, the ranks of P_L and $1-P_L$ are respectively $4L^{-3}n_k$ and $4(1-L^{-3})n_k$. The explicit formula for $\mathcal{B}_{k,y}$ therefore gives

$$-i\int_0^Y \mathrm{d}y\ \mathrm{Tr}[\gamma_5 \mathcal{B}_{k,y}] = -4n_k \log Y + c_k^d n_k + o(1) \quad (Y\to\infty).$$

Insert (54) in the first scalar integral. The terms $4n_k \log Y$ cancel. The remaining phase is $2\pi i n_k$, which disappears on exponentiation, and this proves (74). The integral converges absolutely: block compression bounds $\mathcal{Q}_k S_y \mathcal{Q}_k^\sharp$ uniformly, while each row and column sum of $\mathcal{B}_{k,y}$ is at most $(1+2\exp(1))(b_k^2+y^2)^{-1/2}$. Its trace is bounded by

$$128\exp(1)(1+2\exp(1))^2 a' n_k (b_k^2+y^2)^{-1}.$$

At level zero the spectral identities already give absolute convergence. The uniform kernel bounds below will give a bound independent of the eigenvalues.

6.2. **Local determinant terms.** Choose $\gamma_{\mathrm{w}} \le \gamma$ and $\kappa_{\mathrm{w}} = \kappa + \kappa_0 + 3$ with $\kappa, \kappa_0 > 0$. The propagator bounds above give a fixed constant C_w such that, in component sup norms, for $y \ge 0$,

$$\|\mathbf{1}_{\Delta_z} S_{k,y}(X,A) \mathbf{1}_{\Delta_{z'}}\| \le C_w e^{-\gamma_{\mathrm{w}} d(z,z') - \kappa_{\mathrm{w}} d_M(X)} \quad (k \ge 1),$$
$$|S_{0,y}(X,A)(x,x')|_c \le C_w (1+y)^{-1} e^{-\gamma_{\mathrm{w}} d(c,c') - \kappa_{\mathrm{w}} d_M(X)}.$$

Here $|\cdot|_c$ is the matrix norm induced by the component sup norm. We may use the same C_w after reducing the rates from γ, κ'. For $X \in \mathcal{P}_k$ define

$$\widetilde{E}_k(X,A) = -i b_k^2 \int_0^\infty \mathrm{d}y \ \mathrm{Tr}\big[\gamma_5 \mathcal{B}_{k,y} \mathcal{Q}_k S_{k,y}(X,A) \mathcal{Q}_k^\sharp \mathcal{B}_{k,y}\big] \quad (k \ge 1), \tag{76}$$
$$\widetilde{E}_0(X,A) = \int_1^\infty \frac{\mathrm{d}y}{y} \mathrm{Tr}[F_0 S_{0,y}(X,A)] - i \int_0^1 \mathrm{d}y \ \mathrm{Tr}[\gamma_5 S_{0,y}(X,A)],$$
$$E_k^d(X,A) = \tfrac{1}{4}\big(\widetilde{E}_k(X,A) + \widetilde{E}_k(X,-A) + \overline{\widetilde{E}_k(X,\overline{A})} + \overline{\widetilde{E}_k(X,-\overline{A})}\big),$$
$$E_k^{\det}(X,A) = E_k^d(X,A) - E_k^d(X,0).$$

Set $c_X = 54$ and

$$C_d = 16 c_X C_w \max\left\{\tfrac{\pi}{2} b \exp(1)(1+2\exp(1)),\ 2 + 58\exp(1) + b\exp(1)\right\}.$$

Proposition 6.1. The integrals (76) converge absolutely for $A \in \widetilde{\mathcal{G}}_k$. They define analytic functions, depending only on bonds meeting X, with

$$|\widetilde{E}_k(X,A)|, |E_k^d(X,A)| \le C_d M^3 e^{-(\kappa+1) d_M(X)}. \tag{77}$$

They satisfy

$$\delta Z_k(A) = e^{c_k^d n_k + \sum_X \widetilde{E}_k(X,A)}, \qquad \frac{\delta Z_k(A)}{\delta Z_k(0)} = e^{\sum_X E_k^{\det}(X,A)}. \tag{78}$$

The functions E_k^d are gauge invariant, even in A, and satisfy $\overline{E_k^d(X,\overline{A})} = E_k^d(X,A)$. They are covariant under the lattice step group. ∎

Proof. For $k \ge 1$ write $N_y(X) = \mathcal{Q}_k S_{k,y}(X,A) \mathcal{Q}_k^\sharp$. It is supported on the unit sites of X, and each entry is bounded by $\exp(1) C_w e^{-\kappa_{\mathrm{w}} d_M(X)}$ by normalized block compression. To estimate the trace, cycle one factor $\mathcal{B}_{k,y}$ to the left and use

$$G_y = \mathcal{B}_{k,y} \gamma_5 \mathcal{B}_{k,y} = \gamma_5 \big((b_k + i\gamma_5 y)^{-2}(1 - P_L) + (b_k + b_L + i\gamma_5 y)^{-2} P_L\big).$$

Its column sum in Euclidean matrix norm is at most $(1 + 2\exp(1))/(b_k^2 + y^2)$: the identity contributes one and the two projection coefficients contribute at most $2\exp(1)$. Since $|\mathrm{tr}(mm')| \le 8|m|_2 \max_{i,j} |m'_{ij}|$ for four-component matrices, the integrand is bounded by

$$8\exp(1)(1+2\exp(1)) C_w M^3 |X|_M e^{-\kappa_{\mathrm{w}} d_M(X)} (b_k^2 + y^2)^{-1}. \tag{79}$$

Integrating and multiplying by b_k^2 leaves the factor $\pi b_k/2 \le \pi b/2$.

At $k = 0$, the formula for D_A and the averaging penalty give the scalar component row bound

$$\sup_{x,\beta} \sum_{\beta'} \int \mathrm{d}x' \, |F_0(A)(x,x')_{\beta\beta'}| \le R_F := 58\exp(1) + 1 + b\exp(1). \tag{80}$$

The diagonal and three internal hops contribute at most $41 + \mu$, the external hop at most $6e^{\mathrm{i}(A)}$, and the block penalty at most $b\exp(1)$. The bound follows from $\mu \le 1$ and $e^{\mathrm{i}(A)} \le \exp(1)$. Point evaluation has normalized kernel $48\mathbf{1}[x = x']$; its factor 48 cancels the chamber measure. The two traces in the level-zero integral are consequently bounded by

$$4 R_F C_w M^3 |X|_M (1+y)^{-1} e^{-\kappa_{\mathrm{w}} d_M(X)}, \qquad 4 C_w M^3 |X|_M (1+y)^{-1} e^{-\kappa_{\mathrm{w}} d_M(X)}.$$

The second bound uses that γ_5 has four entries of modulus one. Their integrals contribute respectively $\int_1^\infty (y(1+y))^{-1} \mathrm{d}y = \log 2$ and $\int_0^1 (1+y)^{-1} \mathrm{d}y = \log 2$.

The geometric bound (83) gives $|X|_M \le c_X(d_M(X)+1) \le c_X e^{d_M(X)}$. Applying this to the trace bounds, and using $\kappa_{\mathrm{w}} - 1 \ge \kappa + 1$, proves (77) for $\widetilde{E}_k$ and hence for its four-term mean. The majorants are independent of A and integrable in y. Each integrand is analytic in A; Cauchy's formula on a compact polydisk and the majorant permit its contour integrals to be exchanged with the y integral. The result is analytic.

Every phase in the compressed trace belongs to a unit block or an L-block meeting X, and that block is contained in X by nesting. The fine propagator is supported on X and uses only its connection. At level zero a nonzero external-hop trace uses a bond meeting X; the penalty uses the L-block of its row. This proves locality. Gauge transformations conjugate each product under the trace and hence leave it unchanged. A lattice transformation whose spin matrix commutes with γ_5 does the same. Every spatial action of the step group has such a lift: multiplication by $\gamma_1\gamma_2\gamma_3$ changes its commutation sign with γ_5 and leaves its spatial action fixed. This proves step covariance. The mean in (76) then makes E_k^d even and real in the stated analytic sense.

For real connections, sum the polymer propagators and use (74) or (75). This proves the first identity of (78) there. Both sides are analytic on $\widetilde{\mathcal{G}}_k$, which is open and star-shaped. They agree on a real neighbourhood of zero, so their Taylor series agree there and analytic continuation proves the identity throughout the domain.

To justify the symmetrized quotient, put $\mathcal{L}(A) = \sum_X \widetilde{E}_k(X, A)$. The charge-conjugation transpose and the complex-conjugation identity give $\delta Z_k(-A) = \delta Z_k(A) = \overline{\delta Z_k(\overline{A})}$. For the latter, the spin matrix $\varkappa = \gamma_5 \mathsf{C}$ conjugates $\overline{K_k(\overline{A})}$ to $K_k(-A)$. Thus $\mathcal{L}(-A) - \mathcal{L}(A)$ and $\overline{\mathcal{L}(\overline{A})} - \mathcal{L}(A)$ take values in $2\pi i\mathbb{Z}$. They are continuous on the connected domain, so they are constant. Subtracting their values at zero gives

$$\sum_X (E_k^d(X, A) - E_k^d(X, 0)) = \mathcal{L}(A) - \mathcal{L}(0).$$

This proves the second identity in (78) without a choice of logarithm branch for the complex determinant. □

6.3. **The small-field gain for the determinant.** For a photon connection write $A_\mu(c) = A(c, c + \eta_k e_\mu)$ and

$$\partial_\nu A_\mu(c) = \eta_k^{-1}(A_\mu(c + \eta_k e_\nu) - A_\mu(c)), \qquad \|\delta_\alpha \partial A\|_\infty = \sup_{\mu,\nu} \sup_{0<d(c,c')\le 1} \frac{|\partial_\nu A_\mu(c') - \partial_\nu A_\mu(c)|}{d(c, c')^\alpha}.$$

Local versions use only differences and bonds inside the given set. In this section the curl is mesh normalized: $(dA)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Let R_k be the complex bond functions whose three norms $\|A\|_\infty$, $\|\partial A\|_\infty$, and $\|\delta_\alpha \partial A\|_\infty$ are less than $e_k^{-3/4+\varepsilon}$, $e_k^{-3/4+2\varepsilon}$, and $e_k^{-3/4+3\varepsilon}$, respectively. Let $\widetilde{R}_k$ consist of $A_0 + A_1$, with $A_1 \in R_k$ and A_0 real, such that on every $\square^\dagger = \square^{\sim L}$ one can write $A_0 = A' + d\lambda$, where λ, A' are real and the same three local bounds hold for A'. The external connection domain is

$$\mathfrak{X}_k = \{\mathscr{C} : \|\mathscr{C}\|_\infty < e_k^{-1+2\varepsilon}, \quad \|\partial\mathscr{C}\|_\infty < e_k^{-1+3\varepsilon}\}.$$

If $e_k^{1/4} + e_k^\varepsilon < 1$, both $\mathfrak{X}_k$ and $\widetilde{R}_k + \mathfrak{X}_k$ lie in $\widetilde{\mathcal{G}}_k$: on the local cubes the real part has a representative of size $e_k^{-3/4+\varepsilon}$, and the remaining part has size at most $(e_k^{1/4} + e_k^\varepsilon)e_k^{-1+\varepsilon}$.

Put $v_k = -\log e_k$. Assume

$$v_k \ge 9, \qquad e_k^\varepsilon \le \tfrac{1}{4}, \qquad 8M(v_k/4 + 4)e_k^\varepsilon \le 1, \qquad \ell \ge M v_k^2. \tag{81}$$

Proposition 6.2. For $\mathscr{A} \in \widetilde{R}_k$, $\mathscr{C} \in \mathfrak{X}_k$, and $X \in \mathcal{P}_k$, assume (81) and $e_k^{1/4} + e_k^\varepsilon < 1$. Then

$$|E_k^d(X, \mathscr{A} + \mathscr{C}) - E_k^d(X, \mathscr{C})| \le 8C_d M^3 e_k^{1/4} e^{-\kappa d_M(X)}. \tag{82}$$

■

Proof. If $d_M(X) > v_k/4$, bound the two terms separately by (77) and use $e^{-d_M(X)} \le e_k^{1/4}$. Suppose therefore that $d_M(X) \le v_k/4$. The polymer lift and diameter estimates give a nonwrapping lattice box W containing every bond meeting X, with each side of length at most $s_W = M(d_M(X)+3)$. Their applicability follows from $2M(d_M(X)+2) < \ell$ and $M(d_M(X)+4) < \ell$, which hold by (81).

Write $\mathscr{A} = A_0 + A_1$ as in the domain definition. On each plaquette, the local real representative gives $\|dA_0\|_\infty < 2e_k^{-3/4+2\varepsilon}$. Choose a fixed-order rectilinear path $\Gamma(o,c)$ from a site $o \in W$ and set $\lambda(c) = \eta_k A_0(\Gamma(o,c))$ on W. Then

$$\|A_0 - d\lambda\|_{\infty,W} \le 2s_W \|dA_0\|_{\infty,W} < 4s_W e_k^{-3/4+2\varepsilon}.$$

To see the factor $2s_W$, append one bond to $\Gamma(o,c)$ and subtract $\Gamma(o,c')$. After cancelling common bonds, the resulting loop bounds at most two rectangles of width η_k. Together they contain at most $2s_W/\eta_k$ plaquettes, each contributing $\eta_k dA_0$. The paths and rectangles remain in W because it is a product of intervals. Extend the real function λ by zero outside the projected box. On every bond meeting X, the bound is unchanged. By (81) it is at most $\frac{1}{2}e_k^{-3/4+\varepsilon}$.

Let $B = A_0 - d\lambda + A_1$ on bonds meeting X and zero elsewhere. Then $\|B\|_\infty < \frac{3}{2}e_k^{-3/4+\varepsilon}$. For $|t| \le \frac{1}{3}e_k^{-1/4}$,

$$\|tB + \mathscr{C}\|_\infty < (\tfrac{1}{2} + e_k^{\varepsilon})e_k^{-1+\varepsilon} \le \tfrac{3}{4}e_k^{-1+\varepsilon}.$$

Thus this connection belongs to the complex connection ball $\mathcal{G}_k = \{A : \|A\|_\infty < e_k^{-1+\varepsilon}\}$, which is contained in $\widetilde{\mathcal{G}}_k$, and $f(t) = E_k^d(X, tB + \mathscr{C})$ is analytic on this disk and bounded by the right side of (77). Gauge invariance and locality identify the difference in (82) with $f(1) - f(0)$. On the circle of radius $r = \frac{1}{4}e_k^{-1/4} > 2$, Cauchy's formula yields

$$f(1) - f(0) = \frac{1}{2\pi i}\oint_{|t|=r} \frac{f(t)}{t(t-1)}\,\mathrm{d}t, \qquad |f(1) - f(0)| \le \frac{\sup|f|}{r-1} \le 8e_k^{1/4}\sup|f|.$$

This proves the claimed gain, retaining the stronger decay $e^{-(\kappa+1)d_M(X)}$ in this case. □

Proof of Theorem B. Proposition 4.2 and (56) give invertibility on the two integration spaces. The weighted completion calculation gives (71), with the first-step weight from (70). Proposition 6.1 gives the analytic determinant terms, their bound, symmetries and normalized quotient. □

Appendix A. Polymer geometry

We use the geometric arguments of [Cur, Appendix A]. Let L be odd, M a power of L, and ℓ a multiple of LM with $\ell \ge 13LM$. The M-cubes have centres in $M\mathbb{Z}^3/\ell\mathbb{Z}^3$. They partition the lattice sites; their closed continuum versions may meet along faces and edges. A polymer is a nonempty union of M-cubes connected through faces. Denote its number of cubes by $|X|_M$. A net in the continuum torus is a connected finite union of images of specified closed straight segments in $\mathbb{R}^3$, a point being allowed as a segment. Its length is the sum of the Euclidean lengths of those segments. Set

$$Md_M(X) = \inf\{\text{length}(N) : N \text{ is a net meeting every closed cube of } X\},$$

where the infimum also extends over the finite segment representations. The same definition is used for finite face-connected unions of M-cubes in $\mathbb{R}^3$, with nets in $\mathbb{R}^3$. Distances are sup distances.

A.1. **Size and diameter.** The following bounds hold on the torus and on the covering space:

$$d_M(X) \le |X|_M - 1, \qquad |X|_M \le 54(d_M(X)+1), \tag{83}$$

$$d(p,q) \le M(d_M(X)+2) \qquad (p,q \text{ in the closed cubes of } X). \tag{84}$$

To prove the first inequality of (83), order the cubes along a spanning tree of their adjacency graph. A net meeting one cube can be extended to an adjacent cube by a segment of length at most

M perpendicular to their common face. Starting with one point and repeating this construction gives a net of length at most $M(|X|_M - 1)$.

For the reverse inequality, subdivide the segments of any net N at their intersections and discard overlapping pieces. Its total length does not increase. Doubling the edges of the resulting connected graph gives a closed traversal of length at most $2\,\mathrm{length}(N)$. Cut it into at most $2\,\mathrm{length}(N)/M + 1$ arcs of length at most M. Each arc lifts to a curve in $\mathbb{R}^3$ whose range in each coordinate has length at most M. Such an interval meets at most three closed cube intervals, so an arc meets at most 27 closed M-cubes. Every cube of X meets one of these arcs. Taking the infimum over N gives

$$|X|_M \le 27(2d_M(X) + 1) \le c_X(d_M(X) + 1).$$

For (84), take a net of length less than $Md_M(X) + \delta$ and points p', q' of the net in the respective cubes of p, q. The path in the net from p' to q' has length at most that of the net. Hence $d(p, q) \le M + (Md_M(X) + \delta) + M$; let $\delta \downarrow 0$. The same argument holds in $\mathbb{R}^3$.

A.2. **Nonwrapping lifts.** Let $\mathrm{pr} : \mathbb{R}^3 \to \mathbb{R}^3/\ell\mathbb{Z}^3$ be the projection.

Proposition A.1. If X is a polymer and

$$2M(d_M(X) + 2) < \ell, \tag{85}$$

then $X = \mathrm{pr}(\widetilde{X})$ for a finite face-connected union of M-cubes in $\mathbb{R}^3$. The projection is injective on $\widetilde{X}$, preserves adjacency of its cubes in both directions, and $d_M(\widetilde{X}) = d_M(X)$. The lift is unique up to a translation in $\ell\mathbb{Z}^3$. ■

Proof. Put $D = M(d_M(X) + 2)$, fix p_0 in a closed cube of X, and fix its lift $\widetilde{p}_0$. The projection is injective on the closed sup ball $\widetilde{B} = \{|p - \widetilde{p}_0|_\infty \le D\}$, since two points there differ in every coordinate by at most $2D < \ell$. It maps this ball onto the torus ball $B = \{p : d(p, p_0) \le D\}$. Its inverse $\varphi : B \to \widetilde{B}$ is continuous, and is locally isometric. The diameter estimate places every closed cube of X in B. A segment or a closed M-cube lying in B lifts under φ to a segment of the same length or a closed M-cube: the translates $\widetilde{B} + \ell n$ are pairwise disjoint, and a connected lift lies in one of them. The same inverse maps shared faces to shared faces. Thus $\widetilde{X} = \varphi(X)$ has the asserted injectivity, adjacency, and connectedness.

Any other face-connected injective lift can be translated so that it contains the chosen lift of the cube of p_0. There are no two different neighbours of one lifted cube with the same projection: their centres differ by at most $2M < \ell$. Propagate this observation along a chain of adjacent cubes. Every cube of the other lift equals the corresponding cube of $\widetilde{X}$, proving uniqueness.

Projecting a net for $\widetilde{X}$ gives a net for X with no greater length, so $d_M(X) \le d_M(\widetilde{X})$. Conversely, let N be a net for X of length at most $Md_M(X) + M$, and choose its point q_0 in the cube of p_0. Every point of N is joined to q_0 by a curve of length at most that of N, and $d(q_0, p_0) \le M$. Therefore $N \subset B$. Its lift $\varphi(N)$ has the same length and meets every cube of $\widetilde{X}$. Taking an infimizing sequence of these nets proves the reverse inequality. □

For the determinant estimate, suppose in addition that $M(d_M(X) + 4) < \ell$ and $M \ge 2$. The closed cubes of $\widetilde{X}$ have coordinate ranges of length at most $M(d_M(X) + 2)$ by (84). Enlarge these intervals by one mesh $\eta_k \le 1$ at each end and intersect with $\eta_k\mathbb{Z}^3$. The resulting box W contains every bond meeting the lifted polymer and has each side of length at most $M(d_M(X) + 3)$. This length, increased by one mesh, is less than ℓ. Thus the projection is injective on W, and every torus bond meeting X has exactly one lift in W.

References

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